

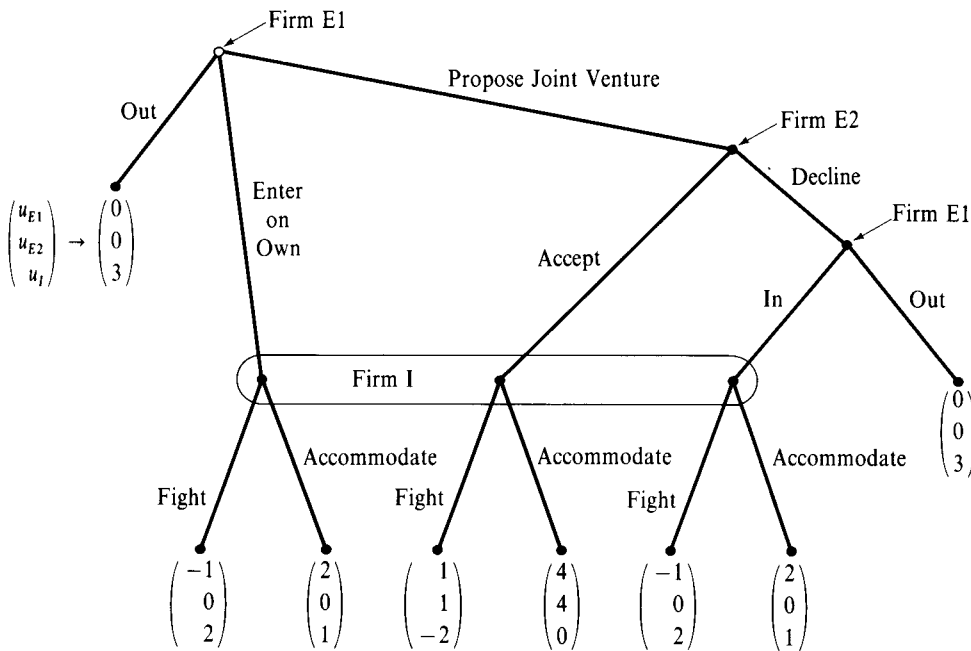


Figure 9.C.2
Extensive form for
Example 9.C.2.

firm E1 must propose the joint venture since if firm E2 will accept its proposal, then firm E1 does better proposing the joint venture than it does by either staying out or entering on its own, regardless of firm I's post-entry strategy. Next, these two conclusions imply that firm I's information set is reached with positive probability (in fact, with certainty) in any weak PBE. Applying Bayesian updating at this information set, we conclude that the beliefs at this information set must assign a probability of 1 to being at the middle node. Given this, in any weak PBE firm I's strategy must be "accommodate if entry occurs." Finally, if firm I is playing "accommodate if entry occurs," then firm E1 must enter if it proposes a joint venture that firm E2 then rejects.

We conclude that the unique weak PBE in this game is a strategy-beliefs pair with strategies of $(\sigma_{E1}, \sigma_{E2}, \sigma_I) = ((\text{propose joint venture, in if E2 declines}), (\text{accept}), (\text{accommodate if entry occurs}))$ and a belief system of μ (middle node of incumbent's information set) = 1. Note that this is not the only Nash equilibrium or, for that matter, the only SPNE. For example, $(\sigma_{E1}, \sigma_{E2}, \sigma_I) = ((\text{out, out if E2 declines}), (\text{decline}), (\text{fight if entry occurs}))$ is an SPNE in this game. ■

Example 9.C.3: In the games of Examples 9.C.1 and 9.C.2 the trick to identifying the weak PBEs consisted of seeing that some player had an optimal strategy that was independent of her beliefs and/or the future play of her opponents. In the game depicted in Figure 9.C.3, however, this is not so for either player. Firm I is now willing to fight if she thinks that firm E has played "in," and the optimal strategy for firm E depends on firm I's behavior (note that $\gamma > -1$).

To solve this game, we look for a *fixed point* at which the behavior generated by beliefs is consistent with these beliefs. We restrict attention to the case where $\gamma > 0$. [Exercise 9.C.2 asks you to determine the set of weak PBEs when $\gamma \in (-1, 0)$.] Let σ_F be the probability that firm I fights after entry, let μ_1 be firm I's belief that

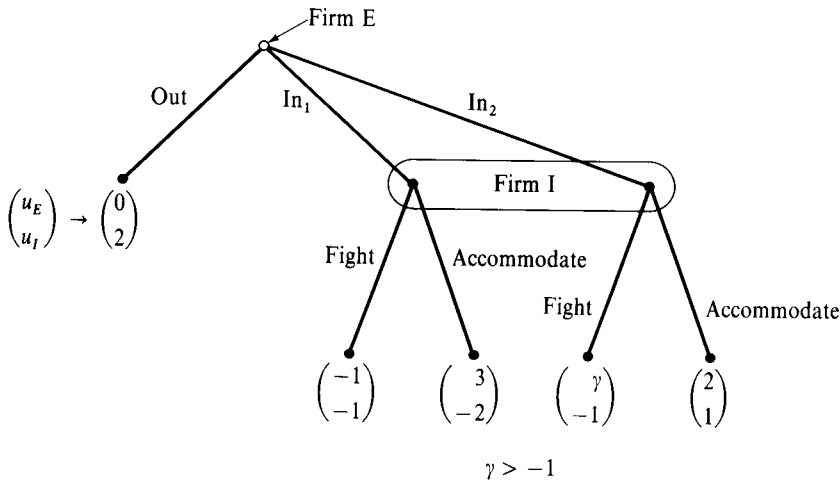


Figure 9.C.3
Extensive form for
Example 9.C.3.

“in₁” was E’s entry strategy if entry has occurred, and let $\sigma_0, \sigma_1, \sigma_2$ denote the probabilities with which firm E actually chooses “out,” “in₁,” and “in₂,” respectively.

Note, first, that firm I is willing to play “fight” with positive probability if and only if $-1 \geq -2\mu_1 + 1(1 - \mu_1)$, or $\mu_1 \geq \frac{2}{3}$.

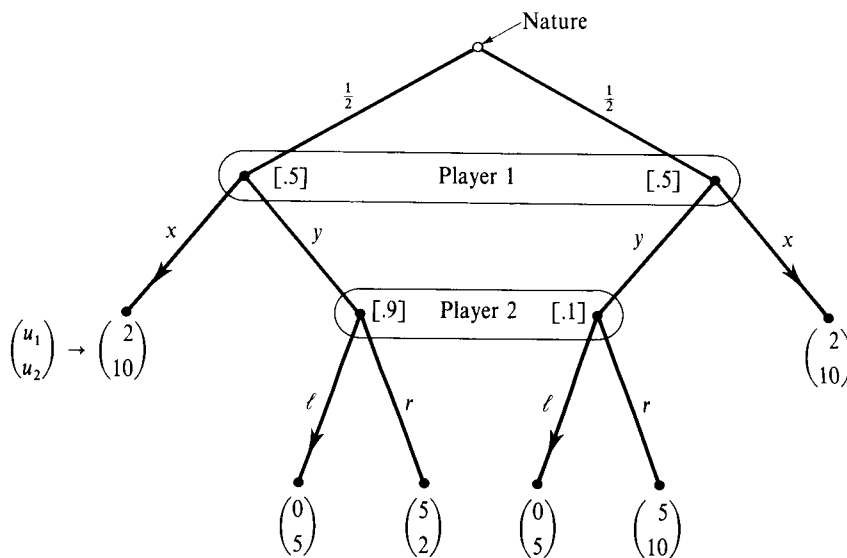
Suppose, first, that $\mu_1 > \frac{2}{3}$ in a weak PBE. Then firm I must be playing “fight” with probability 1. But then firm E must be playing “in₂” with probability 1 (since $\gamma > 0$), and the weak PBE concept would then require that $\mu_1 = 0$, which is a contradiction.

Suppose, instead, that $\mu_1 < \frac{2}{3}$ in a weak PBE. Then firm I must be playing “accommodate” with probability 1. But, if so, then firm E must be playing “in₁” with probability 1, and the weak PBE concept then requires that $\mu_1 = 1$, another contradiction.

Hence, in any weak PBE of this game, we must have $\mu_1 = \frac{2}{3}$. If so, then firm E must be randomizing in the equilibrium with positive probabilities attached to both “in₁” and “in₂” and with “in₁” twice as likely as “in₂.” This means that firm I’s probability of playing “fight” must make firm E indifferent between “in₁” and “in₂.” Hence, we must have $-1\sigma_F + 3(1 - \sigma_F) = \gamma\sigma_F + 2(1 - \sigma_F)$, or $\sigma_F = 1/(\gamma + 2)$. Firm E’s payoff from playing “in₁” or “in₂” is then $(3\gamma + 2)/(\gamma + 2) > 0$, and so firm E must play “out” with zero probability. Therefore, the unique weak PBE in this game when $\gamma > 0$ has $(\sigma_0, \sigma_1, \sigma_2) = (0, \frac{2}{3}, \frac{1}{3})$, $\sigma_F = 1/(\gamma + 2)$, and $\mu_1 = \frac{2}{3}$. ■

Strengthenings of the Weak Perfect Bayesian Equilibrium Concept

We have referred to the concept defined in Definition 9.C.3 as a *weak* perfect Bayesian equilibrium because the consistency requirements that it puts on beliefs are very minimal: The *only* requirement for beliefs, other than that they specify nonnegative probabilities which add to 1 within each information set, is that they are consistent with the equilibrium strategies on the equilibrium path, in the sense of being derived from them through Bayes’ rule. *No restrictions at all are placed on beliefs off the equilibrium path* (i.e., at information sets not reached with positive probability with play of the equilibrium strategies). In the literature, a number of strengthenings of this concept that put additional consistency restrictions on off-the-equilibrium-path

**Figure 9.C.4**

Extensive form for Example 9.C.4. Beliefs in a weak PBE may not be structurally consistent.

beliefs are used. Examples 9.C.4 and 9.C.5 illustrate why a strengthening of the weak PBE concept is often needed.

Example 9.C.4: Consider the game shown in Figure 9.C.4. The pure strategies and beliefs depicted in the figure constitute a weak PBE (the strategies are indicated by arrows on the chosen branches at each information set, and beliefs are indicated by numbers in brackets at the nodes in the information sets). The beliefs satisfy criterion (ii) of Definition 9.C.3; only player 1's information set is reached with positive probability, and player 1's beliefs there do reflect the probabilities assigned by nature. But the beliefs specified for player 2 in this equilibrium are not very sensible; player 2's information set can be reached only if player 1 deviates by instead choosing action y with positive probability, a deviation that must be independent of nature's actual move, since player 1 is ignorant of it. Hence, player 2 could reasonably have only beliefs that assign an equal probability to the two nodes in her information set. Here we see that it is desirable to require that beliefs at least be "structurally consistent" off the equilibrium path in the sense that there is *some* subjective probability distribution over strategy profiles that could generate probabilities consistent with the beliefs. ■

Example 9.C.5: A second and more significant problem is that a weak perfect Bayesian equilibrium need not be subgame perfect. To see this, consider again the entry game in Example 9.B.3. One weak PBE of this game involves strategies of $(\sigma_E, \sigma_I) = ((\text{out}, \text{accommodate if in}), (\text{fight if firm E plays "in"}))$ combined with beliefs for firm I that assign probability 1 to firm E having played "fight." This weak PBE is shown in Figure 9.C.5. But note that these strategies are not subgame perfect; they do not specify a Nash equilibrium in the post-entry subgame.

The problem is that firm I's post-entry belief about firm E's post-entry play is unrestricted by the weak PBE concept because firm I's information set is off the equilibrium path. ■

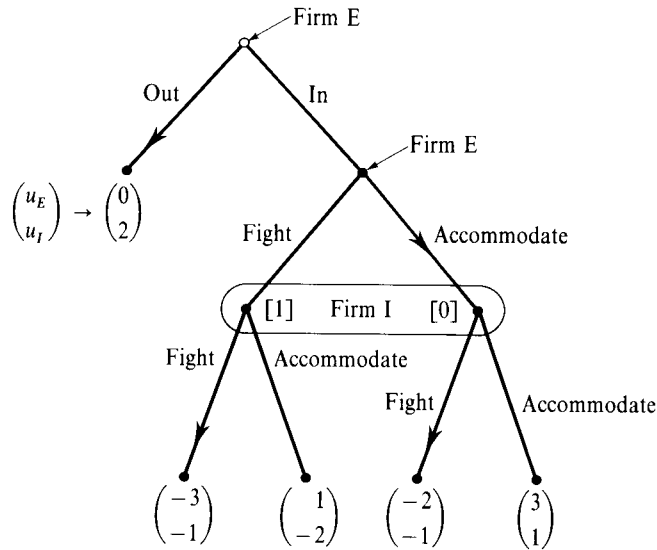


Figure 9.C.5
Extensive form for
Example 9.C.5. A
weak PBE may not be
subgame perfect.

These two examples indicate that the weak PBE concept can be too weak. Thus, in applications in the literature, extra consistency restrictions on beliefs are often added to the weak PBE concept to avoid these problems, with the resulting solution concept referred to as a *perfect Bayesian equilibrium*. (As a simple example, restricting attention to equilibria that induce a weak PBE in every subgame insures subgame perfection.) We shall also do this when necessary later in the book; see, in particular, the discussion of signaling in Section 13.C. For formal definitions and discussion of some notions of perfect Bayesian equilibrium, see Fudenberg and Tirole (1991a) and (1991b).

An important closely related equilibrium notion that also strengthens the weak PBE concept by embodying additional consistency restrictions on beliefs is the *sequential equilibrium* concept developed by Kreps and Wilson (1982). In contrast to notions of perfect Bayesian equilibrium (such as the one we develop in Section 13.C), the sequential equilibrium concept introduces these consistency restrictions indirectly through the formalism of a limiting sequence of strategies. Definition 9.C.4 describes its requirements.

Definition 9.C.4: A strategy profile and system of beliefs (σ, μ) is a *sequential equilibrium* of extensive form game Γ_E if it has the following properties:

- (i) Strategy profile σ is sequentially rational given belief system μ .
- (ii) There exists a sequence of completely mixed strategies $\{\sigma^k\}_{k=1}^\infty$, with $\lim_{k \rightarrow \infty} \sigma^k = \sigma$, such that $\mu = \lim_{k \rightarrow \infty} \mu^k$, where μ^k denotes the beliefs derived from strategy profile σ^k using Bayes' rule.

In essence, the sequential equilibrium notion requires that beliefs be justifiable as coming from some set of totally mixed strategies that are “close to” the equilibrium strategies σ (i.e., a small perturbation of the equilibrium strategies). This can be viewed as requiring that players can (approximately) justify their beliefs by some story in which, with some small probability, players make mistakes in choosing their strategies. Note that every sequential equilibrium is a weak perfect Bayesian equilibrium because the limiting beliefs in Definition 9.C.4 exactly coincide with the beliefs derived from the equilibrium strategies σ via Bayes' rule on the outcome path of strategy profile σ . But, in general, the reverse is not true.

As we now show, the sequential equilibrium concept strengthens the weak perfect Bayesian equilibrium concept in a manner that avoids the problems identified in Examples 9.C.4 and 9.C.5.

Example 9.C.4 Continued: Consider again the game in Figure 9.C.4. In this game, all beliefs that can be derived from any sequence of totally mixed strategies assign equal probability to the two nodes in player 2's information set. Given this fact, in any sequential equilibrium player 2 must play r and player 1 must therefore play y . In fact, strategies (y, r) and beliefs giving equal probability to the two nodes in both players' information sets constitute the unique sequential equilibrium of this game. ■

Example 9.C.5 Continued: The unique sequential equilibrium strategies in the game in Example 9.C.5 (see Figure 9.C.5) are those of the unique SPNE: ((in, accommodate if in), (accommodate if firm E plays "in")). To verify this point, consider any totally mixed strategy $\bar{\sigma}$ and any node x in firm I's information set, which we denote by H_I . Letting z denote firm E's decision node following entry (the initial node of the subgame following entry), the beliefs $\mu_{\bar{\sigma}}$ associated with $\bar{\sigma}$ at information set H_I are equal to

$$\mu_{\bar{\sigma}}(x) = \frac{\text{Prob}(x | \bar{\sigma})}{\text{Prob}(H_I | \bar{\sigma})} = \frac{\text{Prob}(x | z, \bar{\sigma}) \text{Prob}(z | \bar{\sigma})}{\text{Prob}(H_I | z, \bar{\sigma}) \text{Prob}(z | \bar{\sigma})},$$

where $\text{Prob}(x | z, \bar{\sigma})$ is the probability of reaching node x under strategies $\bar{\sigma}$ conditional on having reached node z . Canceling terms and noting that $\text{Prob}(H_I | z, \bar{\sigma}) = 1$, we then have $\mu_{\bar{\sigma}}(x) = \text{Prob}(x | z, \bar{\sigma})$. But this is exactly the probability that firm E plays the action that leads to node x in strategy $\bar{\sigma}$. Thus, any sequence of totally mixed strategies $\{\bar{\sigma}^k\}_{k=1}^{\infty}$ that converge to σ must generate limiting beliefs for firm I that coincide with the play at node z specified in firm E's actual strategy σ_E . It is then immediate that the strategies in any sequential equilibrium must specify Nash equilibrium behavior in this post-entry subgame and thus must constitute a subgame perfect Nash equilibrium. ■

Proposition 9.C.2 gives a general result on the relation between sequential equilibria and subgame perfect Nash equilibria.

Proposition 9.C.2: In every sequential equilibrium (σ, μ) of an extensive form game Γ_E , the equilibrium strategy profile σ constitutes a subgame perfect Nash equilibrium of Γ_E .

Thus, the sequential equilibrium concept strengthens both the SPNE and the weak PBE concepts; every sequential equilibrium is both a weak PBE and an SPNE.

Although the concept of sequential equilibrium restricts beliefs that are off the equilibrium path enough to take care of the problems with the weak PBE concept illustrated in Examples 9.C.4 and 9.C.5, there are some ways in which the requirements on off-equilibrium-path beliefs embodied in the notion of sequential equilibrium may be too strong. For example, they imply that any two players with the same information must have exactly the same beliefs regarding the deviations by other players that have caused play to reach a given part of the game tree.

In Appendix B, we briefly describe another related (and still stronger) solution

concept, an *extensive form trembling-hand perfect Nash equilibrium*, first proposed by Selten (1975).¹⁴

9.D Reasonable Beliefs and Forward Induction

In Section 9.C, we saw the importance of beliefs at unreached information sets for testing the sequential rationality of a strategy. Although the weak perfect Bayesian equilibrium concept and the related stronger concepts discussed in Section 9.C can help rule out noncredible threats, in many games we can nonetheless justify a large range of off-equilibrium-path behavior by picking off-equilibrium-path beliefs appropriately (we shall see some examples shortly). This has led to a considerable amount of recent research aimed at specifying additional restrictions that “reasonable” beliefs should satisfy. In this section, we provide a brief introduction to these ideas. (We shall encounter them again when we study signaling models in Chapter 13, particularly in Appendix A of that chapter.)

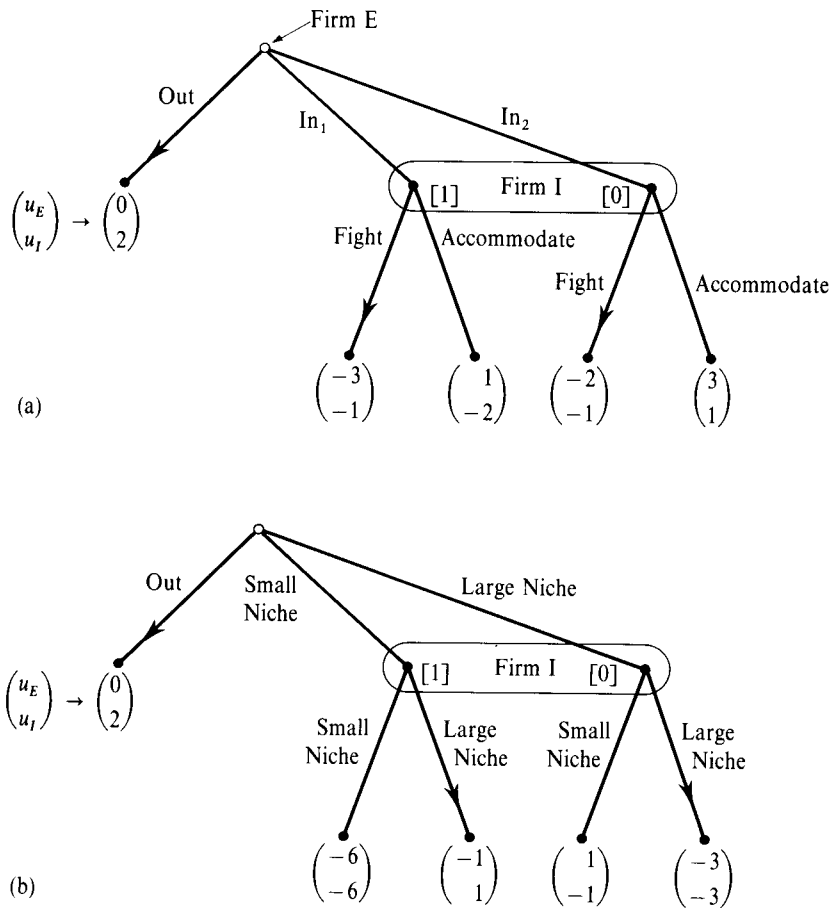
To start, consider the two games depicted in Figure 9.D.1. The first is a variant of the entry game of Figure 9.C.1 in which firm I would now find it worthwhile to fight if it knew that the entrant chose strategy “ in_1 ”; the second is a variant of the Niche Choice game of Example 9.B.4, in which firm E now targets a niche at the time of its entry. Also shown in each diagram is a weak perfect Bayesian equilibrium (arrows denote pure strategy choices, and the numbers in brackets in firm I’s information set denote beliefs).

One can argue that in neither game is the equilibrium depicted very sensible.¹⁵ Consider the game in Figure 9.D.1(a). In the weak PBE depicted, if entry occurs, firm I plays “fight” because it believes that firm E has chosen “ in_1 .” But “ in_1 ” is strictly dominated for firm E by “ in_2 .” Hence, it seems reasonable to think that if firm E decided to enter, it must have used strategy “ in_2 .” Indeed, as is commonly done in this literature, one can imagine firm E making the following speech upon entering: “I have entered, but notice that I would never have used ‘ in_1 ’ to do so because ‘ in_2 ’ is always a better entry strategy for me. Think about this carefully before you choose your strategy.”

A similar argument holds for the weak PBE depicted in Figure 9.D.1(b). Here “small niche” is strictly dominated for firm E, not by “large niche”, but by “out.” Once again, firm I could not reasonably hold the beliefs that are depicted. In this case, firm I should recognize that if firm E entered rather than playing “out,” it must have chosen the large niche. Now you can imagine firm E saying: “Notice that the only way I could ever do better by entering than by choosing ‘out’ is by targeting the large niche.”

14. Selten actually gave it the name *trembling-hand perfect Nash equilibrium*; we add the modifier *extensive form* to help distinguish it from the normal form concept introduced in Section 8.F.

15. For simplicity, we focus on weak perfect Bayesian equilibria here. The points to be made apply as well to the stronger related notions discussed in Section 9.C. In fact, all the weak perfect Bayesian equilibria discussed here are also sequential equilibria; indeed, they are even extensive form trembling-hand perfect.

**Figure 9.D.1**

Two weak PBEs with unreasonable beliefs.

These arguments make use of what is known as *forward induction* reasoning [see Kohlberg (1989) and Kohlberg and Mertens (1986)]. In using backward induction, a player decides what is an optimal action for her at some point in the game tree based on her calculations of the actions that her opponents will rationally play at *later* points of the game. In contrast, in using forward induction, a player reasons about what could have rationally happened *previously*. For example, here firm I decides on its optimal post-entry action by assuming that firm E must have behaved rationally in its entry decision.

This type of idea is sometimes extended to include arguments based on *equilibrium domination*. For example, suppose that we augment the game in Figure 9.D.1(b) by also giving firm I a move after firm E plays "out," as depicted in Figure 9.D.2 (perhaps "out" really involves entry into some alternative market of firm I's in which firm E has only one potential entry strategy).

The figure depicts a weak PBE of this game in which firm E plays "out" and firm I believes that firm E has chosen "small niche" whenever its post-entry information set is reached. In this game, "small niche" is no longer strictly dominated for firm E by "out," so our previous argument does not apply. Nevertheless, if firm E deviates from this equilibrium by entering, we can imagine firm I thinking that since firm E could have received a payoff of 0 by following its equilibrium strategy, it must be hoping to do better than that by entering, and so it must

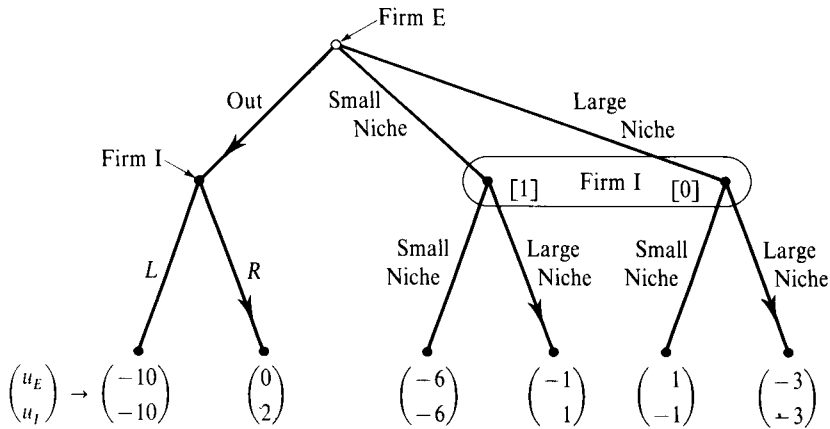


Figure 9.D.2
Strategy “small niche” is equilibrium dominated for firm E.

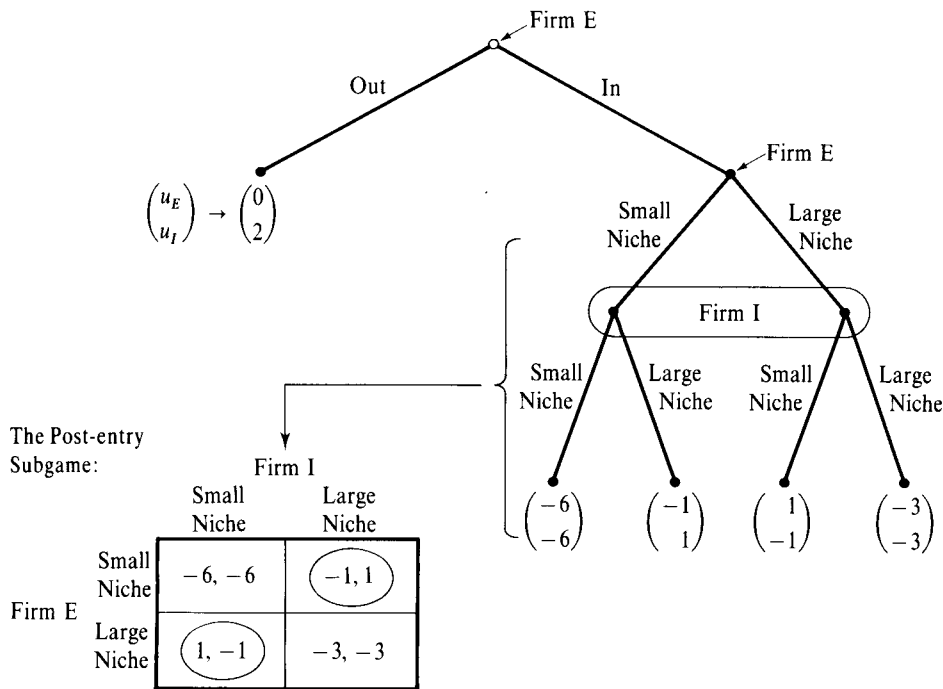


Figure 9.D.3
Forward induction selects equilibrium (large niche, small niche) in the post-entry subgame.

have chosen to target the large niche. In this case, we say that “small niche” is *equilibrium dominated* for firm E; that is, it is dominated if firm E treats its equilibrium payoff as something that it can achieve with certainty by following its equilibrium strategy. (This type of argument is embodied in the *intuitive criterion* refinement that we discuss in Section 13.C and Appendix A of Chapter 13 in the context of signaling models.)

Forward induction can be quite powerful. For example, reconsider the original Niche Choice game depicted in Figure 9.D.3. Recall that there are two (pure strategy) Nash equilibria in the post-entry subgame: (large niche, small niche) and (small niche, large niche). However, the force of the forward induction argument for the game in Figure 9.D.1(b) seems to apply equally well here: Strategy (in, small niche if in) is strictly dominated for firm E by playing “out.” As a result, the incumbent should reason that if firm E has played “in,” it intends to target the large niche in the

post-entry game. If so, firm I is better off targeting the small niche. Thus, forward induction rules out one of the two Nash equilibria in the post-entry subgame.

Although these arguments may seem very appealing, there are also some potential problems. For example, suppose that we are in a world where players make mistakes with some small probability. In such a world, are the forward induction arguments just given convincing? Perhaps not. To see why, suppose that firm E enters in the game shown in Figure 9.D.1(a) when it was supposed to play “out.” Now firm I can explain the deviation to itself as being the result of a mistake on firm E’s part, a mistake that might equally well have led firm E to pick “in₁” as “in₂.” And firm E’s speech may not fall on very sympathetic ears: “Of course, firm E is telling me this,” reasons the incumbent, “it has made a mistake and now is trying to make the best of it by convincing me to accommodate.”

To see this in an even more striking manner, consider the game in Figure 9.D.3. Now, after firm E has entered and the two firms are about to play the simultaneous-move post-entry game, firm E makes its speech. But the incumbent retorts: “Forget it! I think you just made a mistake—and even if you did not, I’m going to target the large niche!”

Clearly, the issues here, although interesting and important, are also tricky.

A noticeable feature of these forward induction arguments is how they use the normal form notion of dominance to restrict predicted play in dynamic games. This stands in sharp contrast with our discussion earlier in this chapter, which relied exclusively on the extensive form to determine how players should play in dynamic games. This raises a natural question: Can we somehow use the normal form representation to predict play in dynamic games?

There are at least two reasons why we might think we can. First, as we discussed in Chapter 7, it seems appealing as a matter of logic to think that players simultaneously choosing their strategies in the normal form (e.g., submitting contingent plans to a referee) is equivalent to their actually playing out the game dynamically as represented in the extensive form. Second, in many circumstances, it seems that the notion of weak dominance can get at the idea of sequential rationality. For example, for finite games of perfect information in which no player has equal payoffs at any two terminal nodes, any strategy profile surviving a process of iterated deletion of weakly dominated strategies leads to the same predicted outcome as the SPNE concept (take a look at Example 9.B.1, and see Exercise 9.D.1).

The argument for using the normal form is also bolstered by the fact that extensive form concepts such as weak PBE can be sensitive to what may seem like irrelevant changes in the extensive form. For example, by breaking up firm E’s decision in the game in Figure 9.D.1(a) into an “out” or “in” decision followed by an “in₁” or “in₂” decision [just as we did in Figure 9.D.3 for the game in Figure 9.D.1(b)], the unique SPNE (and, hence, the unique sequential equilibrium) becomes firm E entering and playing “in₂” and firm I accommodating. However, the reduced normal form associated with these two games (i.e., the normal form where we eliminate all but one of a player’s strategies that have identical payoffs) is invariant to this change in the extensive form; therefore, any solution based on the (reduced) normal form would be unaffected by this change.

These points have led to a renewed interest in the use of the normal form as a device for predicting play in dynamic games [see, in particular, Kohlberg and Mertens (1986)]. At the same time, this issue remains controversial. Many game theorists believe that there is a loss of some information of strategic importance in going from the extensive form to the more condensed normal form. For example, are the games in Figures 9.D.3 and 9.D.1(b) really the same? If you were firm I, would you be as likely to rely on the forward induction argument

in the game in Figure 9.D.3 as in that in Figure 9.D.1(b)? Does it matter for your answer whether in the game in Figure 9.D.3 a minute or a month passes between firm E's two decisions? These issues remain to be sorted out.

APPENDIX A: FINITE AND INFINITE HORIZON BILATERAL BARGAINING

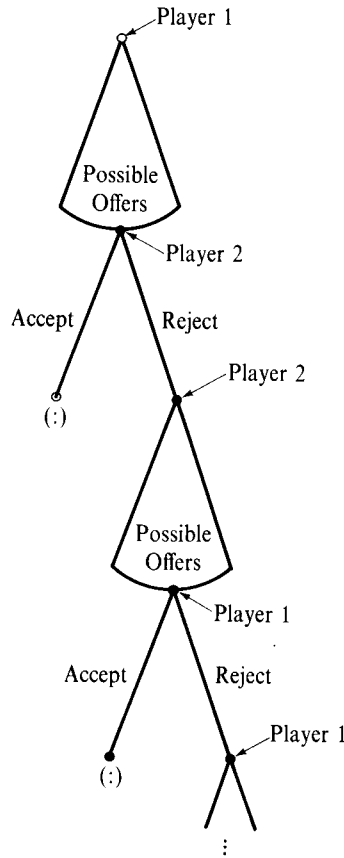
In this appendix we study two models of bilateral bargaining as an economically important example of the use of the subgame perfect Nash equilibrium concept. We begin by studying a finite horizon model of bargaining and then consider its infinite horizon counterpart.

Example 9.AA.1: Finite Horizon Bilateral Bargaining. Two players, 1 and 2, bargain to determine the split of v dollars. The rules are as follows: The game begins in period 1; in period 1, player 1 makes an offer of a split (a real number between 0 and v) to player 2, which player 2 may then accept or reject. If she accepts, the proposed split is immediately implemented and the game ends. If she rejects, nothing happens until period 2. In period 2, the players' roles are reversed, with player 2 making an offer to player 1 and player 1 then being able to accept or reject it. Each player has a discount factor of $\delta \in (0, 1)$, so that a dollar received in period t is worth δ^{t-1} in period 1 dollars. However, after some finite number of periods T , if an agreement has not yet been reached, the bargaining is terminated and the players each receive nothing. A portion of the extensive form of this game is depicted in Figure 9.AA.1 [this model is due to Stahl (1972)].

There is a unique subgame perfect Nash equilibrium (SPNE) in this game. To see this, suppose first that T is odd, so that player 1 makes the offer in period T if no previous agreement has been reached. Now, player 2 is willing to accept *any* offer in this period because she will get zero if she refuses and the game is terminated (she is indifferent about accepting an offer of zero). Given this fact, the unique SPNE in the subgame that begins in the final period when no agreement has been previously reached has player 1 offer player 2 zero and player 2 accept.¹⁶ Therefore, the payoffs from equilibrium play in this subgame are $(\delta^{T-1}v, 0)$.

Now consider play in the subgame starting in period $T-1$ when no previous agreement has been reached. Player 2 makes the offer in this period. In any SPNE, player 1 will accept an offer in period $T-1$ if and only if it provides her with a payoff of at least $\delta^{T-1}v$, since otherwise she will do better rejecting it and waiting to make an offer in period T (she earns $\delta^{T-1}v$ by doing so). Given this fact, in any SPNE, player 2 must make an offer in period $T-1$ that gives player 1 a payoff of exactly $\delta^{T-1}v$, and player 1 accepts this offer (note that this is player 2's best offer

16. Note that if player 2 is unwilling to accept an offer of zero, then player 1 has no optimal strategy; she wants to make a strictly positive offer ever closer to zero (since player 1 will accept any strictly positive offer). If the reliance on player 1 accepting an offer over which she is indifferent bothers you, you can convince yourself that the analysis of the game in which offers must be in small increments (pennies) yields exactly the same outcome as that identified in the text as the size of these increments goes to zero.


Figure 9.AA.1

The alternating-offer bilateral bargaining game.

among all those that would be accepted, and making an offer that will be rejected is worse for player 2 because it results in her receiving a payoff of zero). The payoffs arising if the game reaches period $T - 1$ must therefore be $(\delta^{T-1}v, \delta^{T-2}v - \delta^{T-1}v)$.

Continuing in this fashion, we can determine that the unique SPNE when T is odd results in an agreement being reached in period 1, a payoff for player 1 of

$$\begin{aligned} v_1^*(T) &= v[1 - \delta + \delta^2 - \dots + \delta^{T-1}] \\ &= v \left[(1 - \delta) \left(\frac{1 - \delta^{T-1}}{1 - \delta^2} \right) + \delta^{T-1} \right], \end{aligned}$$

and a payoff to player 2 of $v_2^*(T) = v - v_1^*(T)$.

If T is instead even, then player 1 must earn $v - \delta v_1^*(T - 1)$ because in any SPNE, player 2 (who will be the first offerer in the odd-number-of-periods subgame that begins in period 2 if she rejects player 1's period 1 offer) will accept an offer in period 1 if and only if it gives her at least $\delta v_1^*(T - 1)$, and player 1 will offer her exactly this amount.

Finally, note that as the number of periods grows large ($T \rightarrow \infty$), player 1's payoff converges to $v/(1 + \delta)$, and player 2's payoff converges to $\delta v/(1 + \delta)$. ■

In Example 9.AA.1, the application of the SPNE concept was relatively straightforward; we simply needed to start at the end of the game and work backward. We now consider the infinite horizon counterpart of this game. As we noted in Section

9.B, we can no longer solve for the SPNE in this simple manner when the game has an infinite horizon. Moreover, in many games, introduction of an infinite horizon allows a broad range of behavior to emerge as subgame perfect. Nevertheless, in the infinite horizon bargaining model, the SPNE concept is quite powerful. There is a unique SPNE in this game, and it turns out to be exactly the limiting outcome of the finite horizon model as the length of the horizon T approaches ∞ .

Example 9.AA.2: Infinite Horizon Bilateral Bargaining. Consider an extension of the finite horizon bargaining game considered in Example 9.AA.1 in which bargaining is no longer terminated after T rounds but, rather, can potentially go on forever. If this happens, the players both earn zero. This model is due to Rubinstein (1982).

We claim that this game has a unique SPNE. In this equilibrium, the players reach an immediate agreement in period 1, with player 1 earning $v/(1 + \delta)$ and player 2 earning $\delta v/(1 + \delta)$.

The method of analysis we use here, following Shaked and Sutton (1984), makes heavy use of the stationarity of the game (the subgame starting in period 2 looks exactly like that in period 1, but with the players' roles reversed).

To start, let $\bar{v}_1$ denote the largest payoff that player 1 gets in *any* SPNE (i.e., there may, in principle, be multiple SPNEs in this model).¹⁷ Given the stationarity of the model, this is also the largest amount that player 2 can expect in the subgame that begins in period 2 after her rejection of player 1's period 1 offer, a subgame in which player 2 has the role of being the first player to make an offer. As a result, player 1's payoff in any SPNE cannot be lower than the amount $\underline{v}_1 = v - \delta \bar{v}_1$ because, if it was, then player 1 could do better by making a period 1 offer that gives player 2 just slightly more than $\delta \bar{v}_1$. Player 2 is certain to accept any such offer because she will earn only $\delta \bar{v}_1$ by rejecting it (note that we are using subgame perfection here, because we are requiring that the continuation of play after rejection is an SPNE in the continuation subgame and that player 2's response will be optimal given this fact).

Next, we claim that, in any SPNE, $\bar{v}_1$ cannot be larger than $v - \delta \underline{v}_1$. To see this, note that in any SPNE, player 2 is certain to reject any offer in period 1 that gives her less than $\delta \underline{v}_1$ because she can earn at least $\delta \underline{v}_1$ by rejecting it and waiting to make an offer in period 2. Thus, player 1 can do no better than $v - \delta \underline{v}_1$ by making an offer that is accepted in period 1. What about by making an offer that is rejected in period 1? Since player 2 must earn at least $\delta \underline{v}_1$ if this happens, and since agreement cannot occur before period 2, player 1 can earn no more than $\delta v - \delta \underline{v}_1$ by doing this. Hence, we have $\bar{v}_1 \leq v - \delta \underline{v}_1$.

Next, note that these derivations imply that

$$\bar{v}_1 \leq v - \delta \underline{v}_1 = (\underline{v}_1 + \delta \bar{v}_1) - \delta \underline{v}_1,$$

so that

$$\bar{v}_1(1 - \delta) \leq \underline{v}_1(1 - \delta).$$

Given the definitions of $\underline{v}_1$ and $\bar{v}_1$, this implies that $\underline{v}_1 = \bar{v}_1$, and so player 1's SPNE payoff is uniquely determined. Denote this payoff by v_1° . Since $v_1^\circ = v - \delta v_1^\circ$, we find that player 1 must earn $v_1^\circ = v/(1 + \delta)$ and player 2 must earn $v_2^\circ = v - v_1^\circ = \delta v/(1 + \delta)$. In addition, recalling the argument in the previous paragraph, we see

17. This maximum can be shown to be well defined, but we will not do so here.

that an agreement will be reached in the first period (player 1 will find it worthwhile to make an offer that player 2 accepts). The SPNE strategies are as follows: A player who has just received an offer accepts it if and only if she is offered at least δv_1^0 , while a player whose turn it is to make an offer offers exactly δv_1^0 to the player receiving the offer.

Note that the equilibrium strategies, outcome, and payoffs are precisely the limit of those in the finite game in Example 9.AA.1 as $T \rightarrow \infty$. ■

The coincidence of the infinite horizon equilibrium with the limit of the finite horizon equilibria in this model is not a general property of infinite horizon games. The discussion of infinitely repeated games in Chapter 12 provides an illustration of this point.

We should also point out that the outcomes of game-theoretic models of bargaining can be quite sensitive to the precise specification of the bargaining process and players' preferences. Exercises 9.B.7 and 9.B.13 provide an illustration.

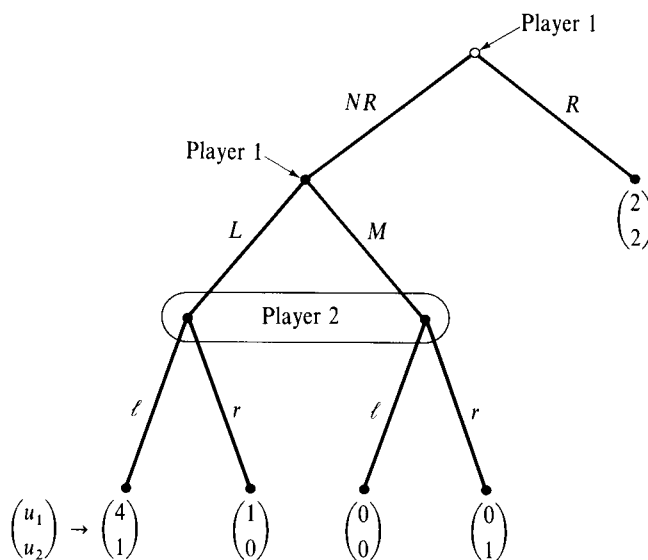
APPENDIX B: EXTENSIVE FORM TREMBLING-HAND PERFECT NASH EQUILIBRIUM

In this appendix we extend the analysis presented in Section 9.C by discussing another equilibrium notion that strengthens the consistency conditions on beliefs in the weak PBE concept: *extensive form trembling-hand perfect Nash equilibrium* [due to Selten (1975)]. In fact, this equilibrium concept is the strongest among those discussed in Section 9.C.

The definition of an extensive form trembling-hand perfect Nash equilibrium parallels that for the normal form (see Section 8.F) but has the trembles applied not to a player's mixed strategies, but rather to the player's choice at each of her information sets. A useful way to view this idea is with what Selten (1975) calls the *agent normal form*. This is the normal form that we would derive if we pretended that the player had a set of agents in charge of moving for her at each of her information sets (a different one for each), each acting independently to try to maximize the player's payoff.

Definition 9.BB.1: Strategy profile σ in extensive form game Γ_E is an *extensive form trembling-hand perfect Nash equilibrium* if and only if it is a normal form trembling-hand perfect Nash equilibrium of the agent normal form derived from Γ_E .

To see why it is desirable to have the trembles occurring at each information set rather than over strategies as in the normal-form concept considered in Section 8.F, consider Figure 9.BB.1, which is taken from van Damme (1983). This game has a unique subgame perfect Nash equilibrium: $(\sigma_1, \sigma_2) = ((NR, L), \ell)$. But you can check that $((NR, L), \ell)$ is not the only normal form trembling-hand perfect Nash equilibrium: so are $((R, L), r)$ and $((R, M), r)$. The reason that these two strategy profiles are normal form trembling-hand perfect is that, in the normal form, the tremble to strategy (NR, M) by player 1 can be larger than that to (NR, L) despite the fact that the latter is a better choice for player 1 at her second decision node.

**Figure 9.BB.1**

Strategy profiles $((R, L), r)$ and $((R, M), r)$ are normal form trembling-hand perfect but are not subgame perfect.

With such a tremble, player 2's best response to player 1's perturbed strategy is r . It is not difficult to see, however, that the unique extensive form trembling-hand perfect Nash equilibrium of this game is $((NR, L), \ell)$ because the agent who moves at player 1's second decision node will put as high a probability as possible on L .

When we compare Definitions 9.BB.1 and 9.C.4, it is apparent that every extensive form trembling-hand perfect Nash equilibrium is a sequential equilibrium. In particular, even though the trembling-hand perfection criterion is not formulated in terms of beliefs, we can use the sequence of (strictly mixed) equilibrium strategies $\{\sigma^k\}_{k=1}^{\infty}$ in the perturbed games of the agent normal form as our strategy sequence for deriving sequential equilibrium beliefs. Because the limiting strategies σ in the extensive form trembling-hand perfect equilibrium are best responses to every element of this sequence, they are also best responses to each other with these derived beliefs. (Every extensive form trembling-hand perfect Nash equilibrium is therefore also subgame perfect.)

In essence, by introducing trembles, the extensive form trembling-hand perfect equilibrium notion makes every part of the tree be reached when strategies are perturbed, and because equilibrium strategies are required to be best responses to perturbed strategies, it insures that equilibrium strategies are sequentially rational. The primary difference between this notion and that of sequential equilibrium is that, like its normal form cousin, the extensive form trembling-hand perfect equilibrium concept can also eliminate some sequential equilibria in which weakly dominated strategies are played. Figure 9.BB.2 (a slight modification of the game in Figure 9.C.1) depicts a sequential equilibrium whose strategies are not extensive form trembling-hand perfect.

In general, however, the concepts are quite close [see Kreps and Wilson (1982) for a formal comparison]; and because it is much easier to check that strategies are best responses at the limiting beliefs than it is to check that they are best responses for a sequence of strategies, sequential equilibrium is much more commonly used. For an interesting further discussion of this concept, consult van Damme (1983).

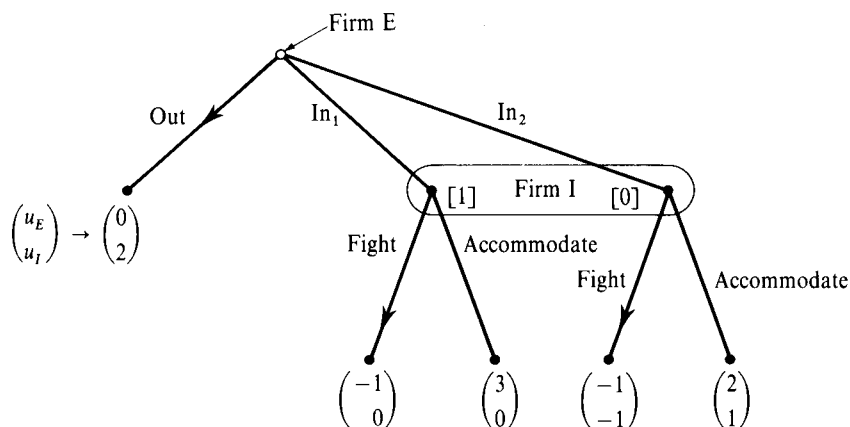


Figure 9.BB.2
A sequential equilibrium need not be extensive form trembling-hand perfect.

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EXERCISES

9.B.1^A How many subgames are there in the game of Example 9.B.2 (depicted in Figure 9.B.3)?

9.B.2^A In text.

9.B.3^B Verify that the strategies identified through backward induction in Example 9.B.2 constitute a Nash equilibrium of the game studied there. Also, identify *all other* pure strategy Nash equilibria of this game. Argue that each of these other equilibria does not satisfy the principle of sequential rationality.

9.B.4^B Prove that in a finite *zero-sum* game of perfect information, there are unique subgame perfect Nash equilibrium payoffs.

9.B.5^B (E. Maskin) Consider a game with two players, player 1 and player 2, in which each player i can choose an action from a finite set M_i that contains m_i actions. Player i 's payoff if the action choices are (m_1, m_2) is $\phi_i(m_1, m_2)$.

(a) Suppose, first, that the two players move simultaneously. How many strategies does each player have?

(b) Now suppose that player 1 moves first and that player 2 observes player 1's move before choosing her move. How many strategies does each player have?

(c) Suppose that the game in (b) has multiple SPNEs. Show that if this is the case, then there exist two pairs of moves (m_1, m_2) and (m'_1, m'_2) (where either $m_1 \neq m'_1$ or $m_2 \neq m'_2$) such that either

$$(i) \quad \phi_1(m_1, m_2) = \phi_1(m'_1, m'_2)$$

or

$$(ii) \quad \phi_2(m_1, m_2) = \phi_2(m'_1, m'_2).$$

(d) Suppose that for any two pairs of moves (m_1, m_2) and (m'_1, m'_2) such that $m_1 \neq m'_1$ or $m_2 \neq m'_2$, condition (ii) is violated (i.e., player 2 is never indifferent between pairs of moves). Suppose also that there exists a pure strategy Nash equilibrium in the game in (a) in which π_1 is player 1's payoff. Show that in any SPNE of the game in (b), player 1's payoff is at least π_1 . Would this conclusion necessarily hold for any Nash equilibrium of the game in (b)?

(e) Show by example that the conclusion in (d) may fail either if condition (ii) holds for some strategy pairs $(m_1, m_2), (m'_1, m'_2)$ with $m_1 \neq m'_1$ or $m_2 \neq m'_2$ or if we replace the phrase *pure strategy Nash equilibrium* with the phrase *mixed strategy Nash equilibrium*.

9.B.6^B Solve for the mixed strategy equilibrium involving actual randomization in the post-entry subgame of the Niche Choice game in Example 9.B.4. Is there an SPNE that induces this behavior in the post-entry subgame? What are the SPNE strategies?

9.B.7^B Consider the finite horizon bilateral bargaining game in Appendix A (Example 9.AA.1); but instead of assuming that players discount future payoffs, assume that it costs $c < v$ to make an offer. (Only the player making an offer incurs this cost, and players who have made offers incur this cost even if no agreement is ultimately reached.) What is the (unique) SPNE of this alternative model? What happens as T approaches ∞ ?

9.B.8^C Prove that every (finite) game Γ_E has a mixed strategy subgame perfect Nash equilibrium.

9.B.9^B Consider a game in which the following simultaneous-move game is played twice:

		Player 2		
		b_1	b_2	b_3
Player 1	a_1	10, 10	2, 12	0, 13
	a_2	12, 2	5, 5	0, 0
	a_3	13, 0	0, 0	1, 1

The players observe the actions chosen in the first play of the game prior to the second play. What are the pure strategy subgame perfect Nash equilibria of this game?

9.B.10^B Reconsider the game in Example 9.B.3, but now change the post-entry game so that when both players choose “accommodate”, instead of receiving the payoffs $(u_E, u_I) = (3, 1)$, the players now must play the following simultaneous-move game:

		Firm I	
		ℓ	r
Firm E	U	3, 1	0, 0
	D	0, 0	x , 3

What are the SPNEs of this game when $x \geq 0$? When $x < 0$?

9.B.11^B Two firms, A and B, are in a market that is declining in size. The game starts in period 0, and the firms can compete in periods 0, 1, 2, 3, ... (i.e., indefinitely) if they so choose. Duopoly profits in period t for firm A are equal to $105 - 10t$, and they are $10.5 - t$ for firm B. Monopoly profits (those if a firm is the only one left in the market) are $510 - 25t$ for firm A and $51 - 2t$ for firm B.

Suppose that at the start of each period, each firm must decide either to “stay in” or “exit” if it is still active (they do so simultaneously if both are still active). Once a firm exits, it is out of the market forever and earns zero in each period thereafter. Firms maximize their (undiscounted) sum of profits.

What is this game’s subgame perfect Nash equilibrium outcome (and what are the firms’ strategies in the equilibrium)?

9.B.12^C Consider the infinite horizon bilateral bargaining model of Appendix A (Example 9.AA.2). Suppose the discount factors δ_1 and δ_2 of the two players differ. Now what is the (unique) subgame perfect Nash equilibrium?

9.B.13^B What are the subgame perfect Nash equilibria of the infinite horizon version of Exercise 9.B.7?

9.B.14^B At time 0, an incumbent firm (firm I) is already in the widget market, and a potential entrant (firm E) is considering entry. In order to enter, firm E must incur a cost of $K > 0$. Firm E’s only opportunity to enter is at time 0. There are three production periods. In any period in which both firms are active in the market, the game in Figure 9.Ex.1 is played. Firm E moves first, deciding whether to stay in or exit the market. If it stays in, firm I decides whether to fight (the upper payoff is for firm E). Once firm E plays “out,” it is out of

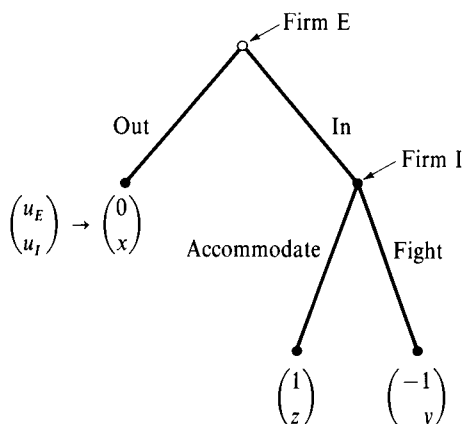


Figure 9.Ex.1

the market forever; firm E earns zero in any period during which it is out of the market, and firm I earns x . The discount factor for both firms is δ .

Assume that:

$$(A.1) \quad x > z > y.$$

$$(A.2) \quad y + \delta x > (1 + \delta)z.$$

$$(A.3) \quad 1 + \delta > K.$$

(a) What is the (unique) subgame perfect Nash equilibrium of this game?

(b) Suppose now that firm E faces a financial constraint. In particular, if firm I fights *once* against firm E (in any period), firm E will be forced out of the market from that point on. Now what is the (unique) subgame perfect Nash equilibrium of this game? (If the answer depends on the values of parameters beyond the three assumptions, indicate how.)

9.C.1^B Prove Proposition 9.C.1.

9.C.2^B What is the set of weak PBEs in the game in Example 9.C.3 when $\gamma \in (-1, 0)$?

9.C.3^C A buyer and a seller are bargaining. The seller owns an object for which the buyer has value $v > 0$ (the seller's value is zero). This value is known to the buyer but not to the seller. The value's prior distribution is common knowledge. There are two periods of bargaining. The seller makes a take-it-or-leave-it offer (i.e., names a price) at the start of each period that the buyer may accept or reject. The game ends when an offer is accepted or after two periods, whichever comes first. Both players discount period 2 payoffs with a discount factor of $\delta \in (0, 1)$.

Assume throughout that the buyer always accepts the seller's offer whenever she is indifferent.

(a) Characterize the (pure strategy) weak perfect Bayesian equilibria for a case in which v can take two values v_L and v_H , with $v_H > v_L > 0$, and where $\lambda = \text{Prob}(v_H)$.

(b) Do the same for the case in which v is uniformly distributed on $[\underline{v}, \bar{v}]$.

9.C.4^C A plaintiff, Ms. P, files a suit against Ms. D (the defendant). If Ms. P wins, she will collect π dollars in damages from Ms. D. Ms. D knows the likelihood that Ms. P will win, $\lambda \in [0, 1]$, but Ms. P does not (Ms. D might know if she was actually at fault). They both have strictly positive costs of going to trial of c_p and c_d . The prior distribution of λ has density $f(\lambda)$ (which is common knowledge).

Suppose pretrial settlement negotiations work as follows: Ms. P makes a take-it-or-leave-it settlement offer (a dollar amount) to Ms. D. If Ms. D accepts, she pays Ms. P and the game is over. If she does not accept, they go to trial.

(a) What are the (pure strategy) weak perfect Bayesian equilibria of this game?

(b) What effects do changes in c_p , c_d , and π have?

(c) Now allow Ms. D, after having her offer rejected, to decide not to go to court after all. What are the weak perfect Bayesian equilibria? What about the effects of the changes in (b)?

9.C.5^C Reconsider Exercise 9.C.4. Now suppose it is Ms. P who knows λ .

9.C.6^B What are the sequential equilibria in the games in Exercises 9.C.3 to 9.C.5?

9.C.7^B (Based on work by K. Bagwell and developed as an exercise by E. Maskin) Consider the extensive form game depicted in Figure 9.Ex.2.

(a) Find a subgame perfect Nash equilibrium of this game. Is it unique? Are there any other Nash equilibria?

(b) Now suppose that player 2 cannot observe player 1's move. Write down the new extensive form. What is the set of Nash equilibria?

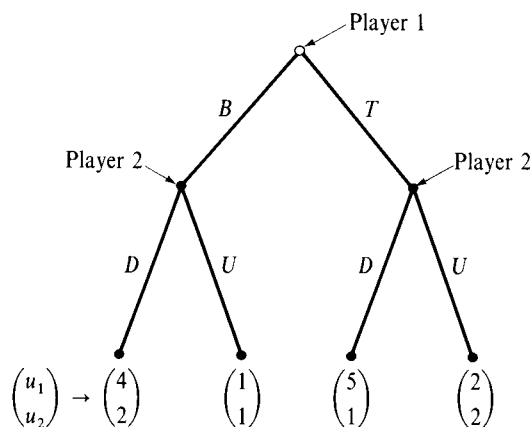
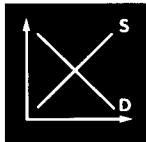


Figure 9.Ex.2

(c) Now suppose that player 2 observes player 1's move correctly with probability $p \in (0, 1)$ and incorrectly with probability $1 - p$ (e.g., if player 1 plays T , player 2 observes T with probability p and observes B with probability $1 - p$). Suppose that player 2's propensity to observe incorrectly (i.e., given by the value of p) is common knowledge to the two players. What is the extensive form now? Show that there is a unique weak perfect Bayesian equilibrium. What is it?

9.D.1^B Show that under the condition given in Proposition 9.B.2 for existence of a unique subgame perfect Nash equilibrium in a finite game of perfect information, there is an order of iterated removal of weakly dominated strategies for which all surviving strategy profiles lead to the same outcome (i.e., have the same equilibrium path and payoffs) as the subgame perfect Nash equilibrium. [In fact, *any* order of deletion leads to this result; see Moulin (1981).]



Market Equilibrium and Market Failure

In Part III, our focus shifts to the fundamental issue of economics: *the organization of production and the allocation of the resulting commodities among consumers*. This fundamental issue can be addressed from two perspectives, one *positive* and the other *normative*.

From a positive (or *descriptive*) perspective, we can investigate the determination of production and consumption under various institutional mechanisms. The institutional arrangement that is our central focus is that of a *market* (or *private ownership*) *economy*. In a market economy, individual consumers have ownership rights to various assets (such as their labor) and are free to trade these assets in the marketplace for other assets or goods. Likewise, firms, which are themselves owned by consumers, decide on their production plan and trade in the market to secure necessary inputs and sell the resulting outputs. Roughly speaking, we can identify a *market equilibrium* as an outcome of a market economy in which each agent in the economy (i.e., each consumer and firm) is doing as well as he can given the actions of all other agents.

In contrast, from a normative (or *prescriptive*) perspective, we can ask what constitutes a *socially optimal* plan of production and consumption (of course, we will need to be more specific about what “socially optimal” means), and we can then examine the extent to which specific institutions, such as a market economy, perform well in this regard.

In Chapter 10, we study *competitive* (or *perfectly competitive*) *market economies* for the first time. These are market economies in which every relevant good is traded in a market at publicly known prices and all agents act as price takers (recall that much of the analysis of individual behavior in Part I was geared to this case). We begin by defining, in a general way, two key concepts: *competitive* (or *Walrasian*) *equilibrium* and *Pareto optimality* (or *Pareto efficiency*). The concept of competitive equilibrium provides us with an appropriate notion of market equilibrium for competitive market economies. The concept of Pareto optimality offers a minimal and uncontroversial test that any social optimal economic outcome should pass. An economic outcome is said to be Pareto optimal if it is impossible to make some individuals better off without making some other individuals worse off. This concept is a formalization of the idea that there is no waste in society, and it conveniently

separates the issue of economic efficiency from more controversial (and political) questions regarding the ideal *distribution* of well-being across individuals.

Chapter 10 then explores these two concepts and the relationships between them in the special context of the *partial equilibrium model*. The partial equilibrium model, which forms the basis for our analysis throughout Part III, offers a considerable analytical simplification; in it, our analysis can be conducted by analyzing a single market (or a small group of related markets) at a time. In this special context, we establish two central results regarding the optimality properties of competitive equilibria, known as the *fundamental theorems of welfare economics*. These can be roughly paraphrased as follows:

The First Fundamental Welfare Theorem. If every relevant good is traded in a market at publicly known prices (i.e., if there is a complete set of markets), and if households and firms act perfectly competitively (i.e., as price takers), then the market outcome is Pareto optimal. That is, when markets are complete, *any competitive equilibrium is necessarily Pareto optimal*.

The Second Fundamental Welfare Theorem. If household preferences and firm production sets are convex, there is a complete set of markets with publicly known prices, and every agent acts as a price taker, then *any Pareto optimal outcome can be achieved as a competitive equilibrium if appropriate lump-sum transfers of wealth are arranged*.

The first welfare theorem provides a set of conditions under which we can be assured that a market economy will achieve a Pareto optimal result; it is, in a sense, the formal expression of Adam Smith's claim about the "invisible hand" of the market. The second welfare theorem goes even further. It states that under the same set of assumptions as the first welfare theorem plus convexity conditions, *all* Pareto optimal outcomes can in principle be implemented through the market mechanism. That is, a public authority who wishes to implement a particular Pareto optimal outcome (reflecting, say, some political consensus on proper distributional goals) may always do so by appropriately redistributing wealth and then "letting the market work."

In an important sense, the first fundamental welfare theorem establishes the perfectly competitive case as a benchmark for thinking about outcomes in market economies. In particular, any inefficiencies that arise in a market economy, and hence any role for Pareto-improving market intervention, *must* be traceable to a violation of at least one of the assumptions of this theorem.

The remainder of Part III, Chapters 11 to 14, can be viewed as a development of this theme. In these chapters, we study a number of ways in which actual markets may depart from this perfectly competitive ideal and where, as a result, market equilibria fail to be Pareto optimal, a situation known as *market failure*.

In Chapter 11, we study *externalities* and *public goods*. In both cases, the actions of one agent directly affect the utility functions or production sets of other agents in the economy. We see there that the presence of these nonmarketed "goods" or "bads" (which violates the complete markets assumption of the first welfare theorem) undermines the Pareto optimality of market equilibrium.

In Chapter 12, we turn to the study of settings in which some agents in the economy have *market power* and, as a result, fail to act as price takers. Once again,

an assumption of the first fundamental welfare theorem fails to hold, and market equilibria fail to be Pareto optimal as a result.

In Chapters 13 and 14, we consider situations in which an *asymmetry of information* exists among market participants. The complete markets assumption of the first welfare theorem implicitly requires that the characteristics of traded commodities be observable by all market participants because, without this observability, distinct markets cannot exist for commodities that have different characteristics. Chapter 13 focuses on the case in which asymmetric information exists between agents at the time of contracting. Our discussion highlights several phenomena—*adverse selection*, *signaling*, and *screening*—that can arise as a result of this informational imperfection, and the welfare loss that it causes. Chapter 14 in contrast, investigates the case of postcontractual asymmetric information, a problem that leads us to the study of the *principal-agent model*. Here, too, the presence of asymmetric information prevents trade of all relevant commodities and can lead market outcomes to be Pareto inefficient.

We rely extensively in some places in Part III on the tools that we developed in Parts I and II. This is particularly true in Chapter 10, where we use material developed in Part I, and Chapters 12 and 13, where we use the game-theoretic tools developed in Part II.

A much more complete and general study of competitive market economies and the fundamental welfare theorems is reserved for Part IV.

Competitive Markets

10.A Introduction

In this chapter, we consider, for the first time, an entire economy in which consumers and firms interact through markets. The chapter has two principal goals: first, to formally introduce and study two key concepts, the notions of *Pareto optimality* and *competitive equilibrium*, and second, to develop a somewhat special but analytically very tractable context for the study of market equilibrium, the *partial equilibrium model*.

We begin in Section 10.B by presenting the notions of a *Pareto optimal* (or *Pareto efficient*) allocation and of a *competitive* (or *Walrasian*) equilibrium in a general setting.

Starting in Section 10.C, we narrow our focus to the partial equilibrium context. The partial equilibrium approach, which originated in Marshall (1920), envisions the market for a single good (or group of goods) for which each consumer's expenditure constitutes only a small portion of his overall budget. When this is so, it is reasonable to assume that changes in the market for this good will leave the prices of all other commodities approximately unaffected and that there will be, in addition, negligible wealth effects in the market under study. We capture these features in the simplest possible way by considering a two-good model in which the expenditure on all commodities other than that under consideration is treated as a single composite commodity (called the *numeraire* commodity), and in which consumers' utility functions take a quasilinear form with respect to this numeraire. Our study of the competitive equilibria of this simple model lends itself to extensive demand-and-supply graphical analysis. We also discuss how to determine the comparative statics effects that arise from exogenous changes in the market environment. As an illustration, we consider the effects on market equilibrium arising from the introduction of a distortionary commodity tax.

In Section 10.D, we analyze the properties of Pareto optimal allocations in the partial equilibrium model. Most significantly, we establish for this special context the validity of the *fundamental theorems of welfare economics*: Competitive equilibrium allocations are necessarily Pareto optimal, and any Pareto optimal allocation can be achieved as a competitive equilibrium if appropriate lump-sum transfers are made.

As we noted in the introduction to Part III, these results identify an important benchmark case in which market equilibria yield desirable economic outcomes. At the same time, they provide a framework for identifying situations of market failure, such as those we study in Chapters 11 to 14.

In Section 10.E, we consider the measurement of welfare changes in the partial equilibrium context. We show that these can be represented by areas between properly defined demand and supply curves. As an application, we examine the deadweight loss of distortionary taxation.

Section 10.F contemplates settings characterized by *free entry*, that is, settings in which all potential firms have access to the most efficient technology and may enter and exit markets in response to the profit opportunities they present. We define a notion of *long-run competitive equilibrium* and then use it to distinguish between long-run and short-run comparative static effects in response to changes in market conditions.

In Section 10.G, we provide a more extended discussion of the use of partial equilibrium analysis in economic modeling.

The material covered in this chapter traces its roots far back in economic thought. An excellent source for further reading is Stigler (1987). We should emphasize that the analysis of competitive equilibrium and Pareto optimality presented here is very much a first pass. In Part IV we return to the topic for a more complete and general investigation; many additional references will be given there.

10.B Pareto Optimality and Competitive Equilibria

In this section, we introduce and discuss the concepts of *Pareto optimality* (or *Pareto efficiency*) and *competitive* (or *Walrasian*) *equilibrium* in a general setting.

Consider an economy consisting of I consumers (indexed by $i = 1, \dots, I$), J firms (indexed by $j = 1, \dots, J$), and L goods (indexed by $\ell = 1, \dots, L$). Consumer i 's preferences over consumption bundles $x_i = (x_{i1}, \dots, x_{iL})$ in his consumption set $X_i \subset \mathbb{R}^L$ are represented by the utility function $u_i(\cdot)$. The total amount of each good $\ell = 1, \dots, L$ initially available in the economy, called the total *endowment* of good ℓ , is denoted by $\omega_\ell \geq 0$ for $\ell = 1, \dots, L$. It is also possible, using the production technologies of the firms, to transform some of the initial endowment of a good into additional amounts of other goods. Each firm j has available to it the production possibilities summarized by the production set $Y_j \subset \mathbb{R}^L$. An element of Y_j is a production vector $y_j = (y_{j1}, \dots, y_{jL}) \in \mathbb{R}^L$. Thus, if $(y_1, \dots, y_J) \in \mathbb{R}^{LJ}$ are the production vectors of the J firms, the total (net) amount of good ℓ available to the economy is $\omega_\ell + \sum_j y_{\ell j}$ (recall that negative entries in a production vector denote input usage; see Section 5.B).

We begin with Definition 10.B.1, which identifies the set of possible outcomes in this economy:

Definition 10.B.1: An *economic allocation* $(x_1, \dots, x_I, y_1, \dots, y_J)$ is a specification of a consumption vector $x_i \in X_i$ for each consumer $i = 1, \dots, I$ and a production vector $y_j \in Y_j$ for each firm $j = 1, \dots, J$. The allocation $(x_1, \dots, x_I, y_1, \dots, y_J)$ is *feasible* if

$$\sum_{i=1}^I x_{\ell i} \leq \omega_\ell + \sum_{j=1}^J y_{\ell j} \quad \text{for } \ell = 1, \dots, L.$$

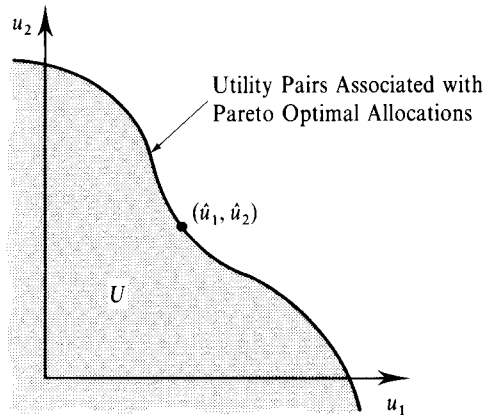


Figure 10.B.1
A utility possibility set.

Thus, an economic allocation is feasible if the total amount of each good consumed does not exceed the total amount available from both the initial endowment and production.

Pareto Optimality

It is often of interest to ask whether an economic system is producing an “optimal” economic outcome. An essential requirement for any optimal economic allocation is that it possess the property of *Pareto optimality* (or *Pareto efficiency*).

Definition 10.B.2: A feasible allocation $(x_1, \dots, x_I, y_1, \dots, y_J)$ is *Pareto optimal* (or *Pareto efficient*) if there is no other feasible allocation $(x'_1, \dots, x'_I, y'_1, \dots, y'_J)$ such that $u_i(x'_i) \geq u_i(x_i)$ for all $i = 1, \dots, I$ and $u_i(x'_i) > u_i(x_i)$ for some i .

An allocation that is Pareto optimal uses society’s initial resources and technological possibilities efficiently in the sense that there is no alternative way to organize the production and distribution of goods that makes some consumer better off without making some other consumer worse off.

Figure 10.B.1 illustrates the concept of Pareto optimality. There we depict the set of attainable utility levels in a two-consumer economy. This set is known as a *utility possibility set* and is defined in this two-consumer case by

$$U = \{(u_1, u_2) \in \mathbb{R}^2 : \text{there exists a feasible allocation } (x_1, x_2, y_1, \dots, y_J) \text{ such that } u_i \leq u_i(x_i) \text{ for } i = 1, 2\}.$$

The set of Pareto optimal allocations corresponds to those allocations that generate utility pairs lying in the utility possibility set’s northeast boundary, such as point $(\hat{u}_1, \hat{u}_2)$. At any such point, it is impossible to make one consumer better off without making the other worse off.

It is important to note that the criterion of Pareto optimality does not insure that an allocation is in any sense equitable. For example, using all of society’s resources and technological capabilities to make a single consumer as well off as possible, subject to all other consumers receiving a subsistence level of utility, results in an allocation that is Pareto optimal but not in one that is very desirable on distributional grounds. Nevertheless, Pareto optimality serves as an important minimal test for the desirability of an allocation; it does, at the very least, say that there is no waste in the allocation of resources in society.

Competitive Equilibria

Throughout this chapter, we are concerned with the analysis of competitive market economies. In such an economy, society's initial endowments and technological possibilities (i.e., the firms) are owned by consumers. We suppose that consumer i initially owns $\omega_{\ell i}$ of good ℓ , where $\sum_i \omega_{\ell i} = \omega_{\ell}$. We denote consumer i 's vector of endowments by $\omega_i = (\omega_{1i}, \dots, \omega_{Li})$. In addition, we suppose that consumer i owns a share θ_{ij} of firm j (where $\sum_i \theta_{ij} = 1$), giving him a claim to fraction θ_{ij} of firm j 's profits.

In a competitive economy, a market exists for each of the L goods, and all consumers and producers act as price takers. The idea behind the price-taking assumption is that if consumers and producers are small relative to the size of the market, they will regard market prices as unaffected by their own actions.¹

Denote the vector of market prices for goods $1, \dots, L$ by $p = (p_1, \dots, p_L)$. Definition 10.B.3 introduces the notion of a competitive (or Walrasian) equilibrium.

Definition 10.B.3: The allocation $(x_1^*, \dots, x_I^*, y_1^*, \dots, y_J^*)$ and price vector $p^* \in \mathbb{R}^L$ constitute a *competitive (or Walrasian) equilibrium* if the following conditions are satisfied:

(i) *Profit maximization:* For each firm j , y_j^* solves

$$\text{Max}_{y_j \in Y_j} p^* \cdot y_j. \quad (10.B.1)$$

(ii) *Utility maximization:* For each consumer i , x_i^* solves

$$\begin{aligned} \text{Max}_{x_i \in X_i} \quad & u_i(x_i) \\ \text{s.t.} \quad & p^* \cdot x_i \leq p^* \cdot \omega_i + \sum_{j=1}^J \theta_{ij}(p^* \cdot y_j^*). \end{aligned} \quad (10.B.2)$$

(iii) *Market clearing:* For each good $\ell = 1, \dots, L$,

$$\sum_{i=1}^I x_{\ell i}^* = \omega_{\ell} + \sum_{j=1}^J y_{\ell j}^*. \quad (10.B.3)$$

Definition 10.B.3 delineates three sorts of conditions that must be met for a competitive economy to be considered to be in equilibrium. Conditions (i) and (ii) reflect the underlying assumption, common to nearly all economic models, that agents in the economy seek to do as well as they can for themselves. Condition (i) states that each firm must choose a production plan that maximizes its profits, taking as given the equilibrium vector of prices of its outputs and inputs (for the justification of the profit-maximization assumption, see Section 5.G). We studied this competitive behavior of the firm extensively in Chapter 5.

Condition (ii) requires that each consumer chooses a consumption bundle that maximizes his utility given the budget constraint imposed by the equilibrium prices and by his wealth. We studied this competitive behavior of the consumer extensively in Chapter 3. One difference here, however, is that the consumer's wealth is now a function of prices. This dependence of wealth on prices arises in

1. Strictly speaking, it is *equilibrium* market prices that they will regard as unaffected by their actions. For more on this point, see the small-type discussion later in this section.

two ways: First, prices determine the value of the consumer's initial endowments; for example, an individual who initially owns real estate is poorer if the price of real estate falls. Second, the equilibrium prices affect firms' profits and hence the value of the consumer's shareholdings.

Condition (iii) is somewhat different. It requires that, at the equilibrium prices, the desired consumption and production levels identified in conditions (i) and (ii) are in fact mutually compatible; that is, the aggregate supply of each commodity (its total endowment plus its net production) equals the aggregate demand for it. If excess supply or demand existed for a good at the going prices, the economy could not be at a point of equilibrium. For example, if there is excess demand for a particular commodity at the existing prices, some consumer who is not receiving as much of the commodity as he desires could do better by offering to pay just slightly more than the going market price and thereby get sellers to offer the commodity to him first. Similarly, if there is excess supply, some seller will find it worthwhile to offer his product at a slight discount from the going market price.²

Note that in justifying why an equilibrium must involve no excess demand or supply, we have actually made use of the fact that consumers and producers *might not* simply take market prices as given. How are we to reconcile this argument with the underlying price-taking assumption?

An answer to this apparent paradox comes from recognizing that consumers and producers *always* have the ability to alter their offered prices (in the absence of any institutional constraints preventing this). For the price-taking assumption to be appropriate, what we want is that they have no *incentive* to alter prices that, if taken as given, equate demand and supply (we have already seen that they *do* have an incentive to alter prices that do not equate demand and supply).

Notice that as long as consumers can make their desired trades at the going market prices, they will not wish to offer more than the market price to entice sellers to sell to them first. Similarly, if producers are able to make their desired sales, they will have no incentive to undercut the market price. Thus, at a price that equates demand and supply, consumers do not wish to raise prices, and firms do not wish to lower them.

More troublesome is the possibility that a buyer might try to lower the price he pays or that a seller might try to raise the price he charges. A seller, for example, may possess the ability to raise profitably prices of the goods he sells above their competitive level (see Chapter 12). In this case, there is no reason to believe that this market power will not be exercised. To rescue the price-taking assumption, one needs to argue that under appropriate (competitive) conditions such market power does not exist. This we do in Sections 12.F and 18.C, where we formalize the idea that if market participants' desired trades are small relative to the size of the market, then they will have little incentive to depart from market prices. Thus, in a suitably defined equilibrium, they will act approximately like price takers.

Note from Definition 10.B.3 that if the allocation $(x_1^*, \dots, x_I^*, y_1^*, \dots, y_J^*)$ and price vector $p^* \gg 0$ constitute a competitive equilibrium, then so do the allocation

2. Strictly speaking, this second part of the argument requires the price to be positive; indeed, if the price is zero (i.e., if the good is free), then excess supply should be permissible at equilibrium. In the remainder of this chapter, however, consumer preferences will be such as to preclude this possibility (goods will be assumed to be desirable). Hence, we neglect this possibility here.

$(x_1^*, \dots, x_I^*, y_1^*, \dots, y_J^*)$ and price vector $\alpha p^* = (\alpha p_1^*, \dots, \alpha p_L^*)$ for any scalar $\alpha > 0$ (see Exercise 10.B.2). As a result, we can normalize prices without loss of generality. In this chapter, we always normalize by setting one good's price equal to 1.

Lemma 10.B.1 will also prove useful in identifying competitive equilibria.

Lemma 10.B.1: If the allocation $(x_1, \dots, x_I, y_1, \dots, y_J)$ and price vector $p \gg 0$ satisfy the market clearing condition (10.B.3) for all goods $\ell \neq k$, and if every consumer's budget constraint is satisfied with equality, so that $p \cdot x_i = p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j$ for all i , then the market for good k also clears.

Proof: Adding up the consumers' budget constraints over the I consumers and rearranging terms, we get

$$\sum_{\ell \neq k} p_\ell \left(\sum_{i=1}^I x_{\ell i} - \omega_\ell - \sum_{j=1}^J y_{\ell j} \right) = -p_k \left(\sum_{i=1}^I x_{ki} - \omega_k - \sum_{j=1}^J y_{kj} \right).$$

By market clearing in goods $\ell \neq k$, the left-hand side of this equation is equal to zero. Thus, the right-hand side must be equal to zero as well. Because $p_k > 0$, this implies that we have market clearing in good k . ■

In the models studied in this chapter, Lemma 10.B.1 will allow us to identify competitive equilibria by checking for market clearing in only $L - 1$ markets. Lemma 10.B.1 is really just a matter of double-entry accountancy. If consumers' budget constraints hold with equality, the dollar value of each consumer's planned purchases equals the dollar value of what he plans to sell plus the dollar value of his share (θ_{ij}) of the firms' (net) supply, and so the total value of planned purchases in the economy must equal the total value of planned sales. If those values are equal to each other in all markets but one, then equality must hold in the remaining market as well.

10.C Partial Equilibrium Competitive Analysis

Marshallian partial equilibrium analysis envisions the market for one good (or several goods, as discussed in Section 10.G) that constitutes a small part of the overall economy. The small size of the market facilitates two important simplifications for the analysis of market equilibrium:³ First, as Marshall (1920) emphasized, when the expenditure on the good under study is a small portion of a consumer's total expenditure, only a small fraction of any additional dollar of wealth will be spent on this good; consequently, we can expect wealth effects for it to be small. Second, with similarly dispersed substitution effects, the small size of the market under study should lead the prices of other goods to be approximately unaffected by changes in this market.⁴ Because of this fixity of other prices, we are justified in treating the expenditure on these other goods as a single composite commodity, which we call the *numeraire* (see Exercise 3.G.5).

3. The following points have been formalized by Vives (1987). (See Exercise 10.C.1 for an illustration.)

4. This is not the only possible justification for taking other goods' prices as being unaffected by the market under study; see Section 10.G.

With this partial equilibrium interpretation as our motivation, we proceed to study a simple two-good quasilinear model. There are two commodities: good ℓ and the numeraire. We let x_i and m_i denote consumer i 's consumption of good ℓ and the numeraire, respectively. Each consumer $i = 1, \dots, I$ has a utility function that takes the quasilinear form (see Sections 3.B and 3.C):

$$u_i(m_i, x_i) = m_i + \phi_i(x_i).$$

We let each consumer's consumption set be $\mathbb{R} \times \mathbb{R}_+$, and so we assume for convenience that consumption of the numeraire commodity m can take negative values. This is to avoid dealing with boundary problems. We assume that $\phi_i(\cdot)$ is bounded above and twice differentiable, with $\phi_i'(x_i) > 0$ and $\phi_i''(x_i) < 0$ at all $x_i \geq 0$. We normalize $\phi_i(0) = 0$.

In terms of our partial equilibrium interpretation, we think of good ℓ as the good whose market is under study and of the numeraire as representing the composite of all other goods (m stands for the total money expenditure on these other goods). Recall that with quasilinear utility functions, wealth effects for non-numeraire commodities are null.

In the discussion that follows, we normalize the price of the numeraire to equal 1, and we let p denote the price of good ℓ .

Each firm $j = 1, \dots, J$ in this two-good economy is able to produce good ℓ from good m . The amount of the numeraire required by firm j to produce $q_j \geq 0$ units of good ℓ is given by the cost function $c_j(q_j)$ (recall that the price of the numeraire is 1). Letting z_j denote firm j 's use of good m as an input, its production set is therefore

$$Y_j = \{(-z_j, q_j): q_j \geq 0 \text{ and } z_j \geq c_j(q_j)\}.$$

In what follows, we assume that $c_j(\cdot)$ is twice differentiable, with $c_j'(q_j) > 0$ and $c_j''(q_j) \geq 0$ at all $q_j \geq 0$. [In terms of our partial equilibrium interpretation, we can think of $c_j(q_j)$ as actually arising from some multiple-input cost function $c_j(\bar{w}, q_j)$, given the fixed vector of factor prices $\bar{w}$.⁵]

For simplicity, we shall assume that there is no initial endowment of good ℓ , so that all amounts consumed must be produced by the firms. Consumer i 's initial endowment of the numeraire is the scalar $\omega_{mi} > 0$, and we let $\omega_m = \sum_i \omega_{mi}$.

We now proceed to identify the competitive equilibria for this two-good quasilinear model. Applying Definition 10.B.3, we consider first the implications of profit and utility maximization.

Given the price p^* for good ℓ , firm j 's equilibrium output level q_j^* must solve

$$\text{Max}_{q_j \geq 0} \quad p^* q_j - c_j(q_j),$$

which has the necessary and sufficient first-order condition

$$p^* \leq c_j'(q_j^*), \quad \text{with equality if } q_j^* > 0.$$

On the other hand, consumer i 's equilibrium consumption vector (m_i^*, x_i^*) must

5. Some of the exercises at the end of the chapter investigate the effects of exogenous changes in these factor prices.

solve

$$\begin{aligned} \text{Max}_{m_i \in \mathbb{R}, x_i \in \mathbb{R}_+} \quad & m_i + \phi_i(x_i) \\ \text{s.t.} \quad & m_i + p^* x_i \leq \omega_{mi} + \sum_{j=1}^J \theta_{ij}(p^* q_j^* - c_j(q_j^*)). \end{aligned}$$

In any solution to this problem, the budget constraint holds with equality. Substituting for m_i from this constraint, we can rewrite consumer i 's problem solely in terms of choosing his optimal consumption of good ℓ . Doing so, we see that x_i^* must solve

$$\text{Max}_{x_i \geq 0} \quad \phi_i(x_i) - p^* x_i + \left[\omega_{mi} + \sum_{j=1}^J \theta_{ij}(p^* q_j^* - c_j(q_j^*)) \right],$$

which has the necessary and sufficient first-order condition

$$\phi_i'(x_i^*) \leq p^*, \quad \text{with equality if } x_i^* > 0.$$

In what follows, it will be convenient to adopt the convention of identifying an equilibrium allocation by the levels of good ℓ consumed and produced, $(x_1^*, \dots, x_I^*, q_1^*, \dots, q_J^*)$, with the understanding that consumer i 's equilibrium consumption of the numeraire is then $m_i^* = [\omega_{mi} + \sum_j \theta_{ij}(p^* q_j^* - c_j(q_j^*))] - p^* x_i^*$ and that firm j 's equilibrium usage of the numeraire as an input is $z_j^* = c_j(q_j^*)$.

To complete the development of the equilibrium conditions for this model, recall that by Lemma 10.B.1, we need only check that the market for good ℓ clears.⁶ Hence, we conclude that the allocation $(x_1^*, \dots, x_I^*, q_1^*, \dots, q_J^*)$ and the price p^* constitute a competitive equilibrium if and only if

$$p^* \leq c_j'(q_j^*), \quad \text{with equality if } q_j^* > 0 \quad j = 1, \dots, J. \quad (10.C.1)$$

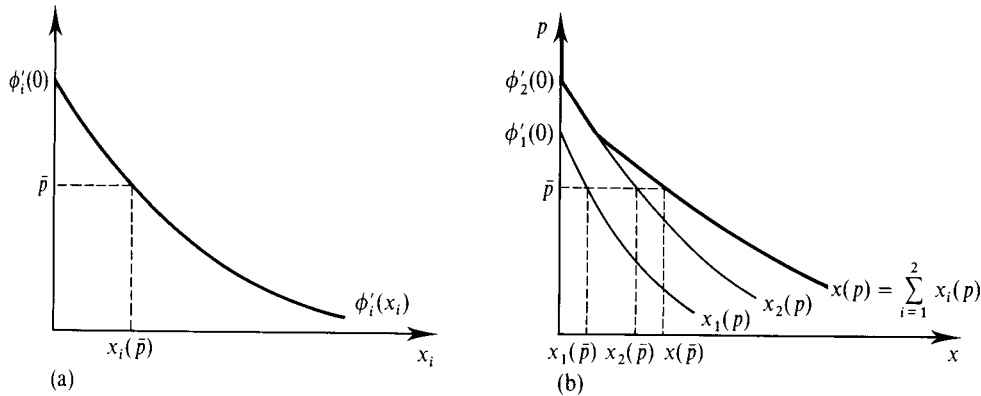
$$\phi_i'(x_i^*) \leq p^*, \quad \text{with equality if } x_i^* > 0 \quad i = 1, \dots, I. \quad (10.C.2)$$

$$\sum_{i=1}^I x_i^* = \sum_{j=1}^J q_j^*. \quad (10.C.3)$$

At any interior solution, condition (10.C.1) says that firm j 's marginal benefit from selling an additional unit of good ℓ , p^* , exactly equals its marginal cost $c_j'(q_j^*)$. Condition (10.C.2) says that consumer i 's marginal benefit from consuming an additional unit of good ℓ , $\phi_i'(x_i^*)$, exactly equals its marginal cost p^* . Condition (10.C.3) is the market-clearing equation. Together, these $I + J + 1$ conditions characterize the $(I + J + 1)$ equilibrium values $(x_1^*, \dots, x_I^*, q_1^*, \dots, q_J^*)$ and p^* . Note that as long as $\text{Max}_i \phi_i'(0) > \text{Min}_j c_j'(0)$, the aggregate consumption and production of good ℓ must be strictly positive in a competitive equilibrium [this follows from conditions (10.C.1) and (10.C.2)]. For simplicity, we assume that this is the case in the discussion that follows.

Conditions (10.C.1) to (10.C.3) have a very important property: They do not involve, in any manner, the endowments or the ownership shares of the consumers. As a result, we see that *the equilibrium allocation and price are independent of the*

6. Note that we must have $p^* > 0$ in any competitive equilibrium; otherwise, consumers would demand an infinite amount of good ℓ [recall that $\phi_i'(\cdot) > 0$].

**Figure 10.C.1**

Construction of the aggregate demand function.
 (a) Determination of consumer i 's demand.
 (b) Construction of the aggregate demand function ($I = 2$).

distribution of endowments and ownership shares. This important simplification arises from the quasilinear form of consumer preferences.⁷

The competitive equilibrium of this model can be nicely represented using the traditional Marshallian graphical technique that identifies the equilibrium price as the point of intersection of aggregate demand and aggregate supply curves.

We can derive the aggregate demand function for good ℓ from condition (10.C.2). Because $\phi'_i(\cdot) < 0$ and $\phi_i(\cdot)$ is bounded, $\phi'_i(\cdot)$ is a strictly decreasing function of x_i taking all values in the set $(0, \phi'_i(0)]$. Therefore, for each possible level of $p > 0$, we can solve for a unique level of x_i , denoted $x_i(p)$, that satisfies condition (10.C.2). Note that if $p \geq \phi'_i(0)$, then $x_i(p) = 0$. Figure 10.C.1(a) depicts this construction for a price $\bar{p} > 0$. The function $x_i(\cdot)$ is consumer i 's *Walrasian demand function* for good ℓ (see Section 3.D) which, because of quasilinearity, does not depend on the consumer's wealth. It is continuous and nonincreasing in p at all $p > 0$, and is strictly decreasing at any $p < \phi'_i(0)$ [at any such p , we have $x'_i(p) = 1/\phi''_i(x_i(p)) < 0$].

The *aggregate demand function for good ℓ* is then the function $x(p) = \sum_i x_i(p)$, which is continuous and nonincreasing at all $p > 0$, and is strictly decreasing at any $p < \max_i \phi'_i(0)$. Its construction is depicted in Figure 10.C.1(b) for the case in which $I = 2$; it is simply the horizontal summation of the individual demand functions and is drawn in the figure with a heavy trace. Note that $x(p) = 0$ whenever $p \geq \max_i \phi'_i(0)$.

The aggregate supply function can be similarly derived from condition (10.C.1).⁸ Suppose, first, that every $c_j(\cdot)$ is strictly convex and that $c'_j(q_j) \rightarrow \infty$ as $q_j \rightarrow \infty$. Then, for any $p > 0$, we can let $q_j(p)$ denote the unique level of q_j that satisfies condition (10.C.1). Note that for $p \leq c'_j(0)$, we have $q_j(p) = 0$. Figure 10.C.2(a) illustrates this construction for a price $\bar{p} > 0$. The function $q_j(\cdot)$ is firm j 's *supply function* for good ℓ (see Sections 5.C and 5.D). It is continuous and nondecreasing at all $p > 0$, and is strictly increasing at any $p > c'_j(0)$ [for any such p , $q'_j(p) = 1/c''_j(q_j(p)) > 0$].

The *aggregate (or industry) supply function* for good ℓ is then the function $q(p) = \sum_j q_j(p)$, which is continuous and nondecreasing at all $p > 0$, and is strictly increasing at any $p > \min_j c'_j(0)$. Its construction is depicted in Figure 10.C.2(b) for

7. See Section 10.G for a further discussion of this general feature of equilibrium in economies with quasilinear utility functions.

8. See Section 5.D for an extensive discussion of individual supply in the one-input, one-output case.

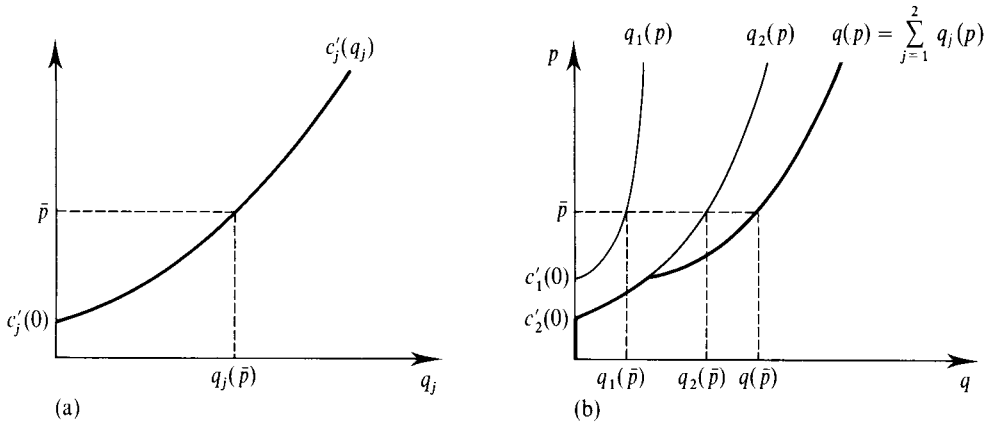


Figure 10.C.2
Construction of the aggregate supply function.
(a) Determination of firm j 's supply.
(b) Construction of the aggregate supply function ($J = 2$).

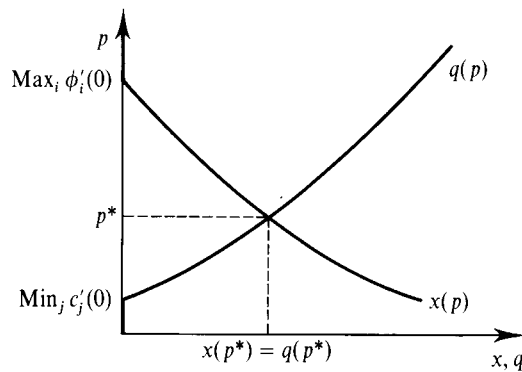


Figure 10.C.3
The equilibrium price equates demand and supply.

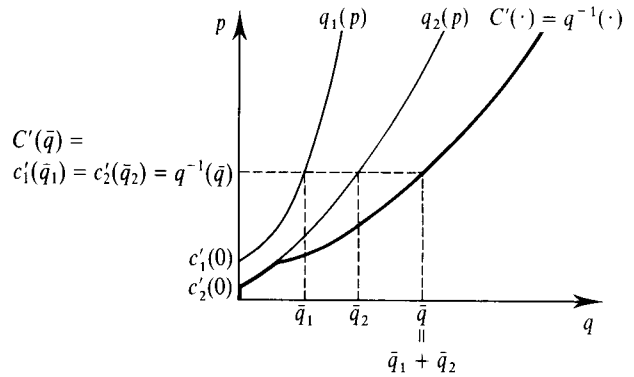
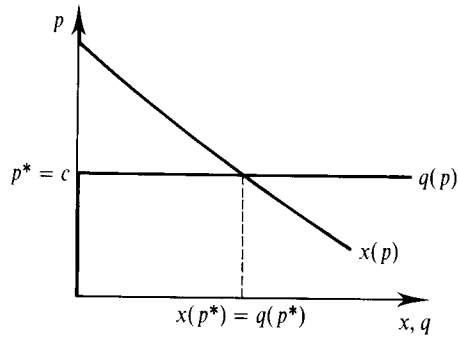
the case in which $J = 2$; it is equal to the horizontal sum of the individual firms' supply functions and is drawn in the figure with a heavy trace. Note that $q(p) = 0$ whenever $p \leq \text{Min}_j c'_j(0)$.

To find the equilibrium price of good ℓ , we need only find the price p^* at which aggregate demand equals aggregate supply, that is, at which $x(p^*) = q(p^*)$. When $\text{Max}_i \phi'_i(0) > \text{Min}_j c'_j(0)$ as we have assumed, at any $p \geq \text{Max}_i \phi'_i(0)$ we have $x(p) = 0$ and $q(p) > 0$. Likewise, at any $p \leq \text{Min}_j c'_j(0)$ we have $x(p) > 0$ and $q(p) = 0$. The existence of an equilibrium price $p^* \in (\text{Min}_j c'_j(0), \text{Max}_i \phi'_i(0))$ then follows from the continuity properties of $x(\cdot)$ and $q(\cdot)$. The solution is depicted in Figure 10.C.3. Note also that because $x(\cdot)$ is strictly decreasing at all $p < \text{Max}_i \phi'_i(0)$ and $q(\cdot)$ is strictly increasing at all $p > \text{Min}_j c'_j(0)$, this equilibrium price is uniquely defined.⁹ The individual consumption and production levels of good ℓ in this equilibrium are then given by $x_i^* = x_i(p^*)$ for $i = 1, \dots, I$ and $q_j^* = q_j(p^*)$ for $j = 1, \dots, J$.

More generally, if some $c_j(\cdot)$ is merely convex [e.g., if $c_j(\cdot)$ is linear, as in the constant returns case], then $q_j(\cdot)$ is a convex-valued correspondence rather than a function and it may be well defined only on a subset of prices.¹⁰ Nevertheless, the

9. Be warned, however, that the uniqueness of equilibrium is a property that need not hold in more general settings in which wealth effects are present. (See Chapter 17.)

10. For example, if firm j has $c_j(q_j) = c_j q_j$ for some scalar $c_j > 0$, then when $p > c_j$, we have $q_j(p) = \infty$. As a result, if $p > c_j$, the aggregate supply is $q(p) = \sum_j q_j(p) = \infty$; consequently $q(\cdot)$ is not well defined for this p .



basic features of the analysis do not change. Figure 10.C.4 depicts the determination of the equilibrium value of p in the case where, for all j , $c_j(q_j) = cq_j$ for some scalar $c > 0$. The only difference from the strictly convex case is that, when $J > 1$, individual firms' equilibrium production levels are not uniquely determined.

The inverses of the aggregate demand and supply functions also have interpretations that are of interest. At any given level of aggregate output of good ℓ , say $\bar{q}$, the inverse of the industry supply function, $q^{-1}(\bar{q})$, gives the price that brings forth aggregate supply $\bar{q}$. That is, when each firm chooses its optimal output level facing the price $p = q^{-1}(\bar{q})$, aggregate supply is exactly $\bar{q}$. Figure 10.C.5 illustrates this point. Note that in selecting these output levels, all active firms set their marginal cost equal to $q^{-1}(\bar{q})$. As a result, the marginal cost of producing an additional unit of good ℓ at $\bar{q}$ is precisely $q^{-1}(\bar{q})$, regardless of which active firm produces it. Thus $q^{-1}(\cdot)$, the inverse of the industry supply function, can be viewed as the *industry marginal cost function*, which we now denote by $C'(\cdot) = q^{-1}(\cdot)$.¹¹

Figure 10.C.4 (left)

Equilibrium when $c_j(q_j) = cq_j$ for all $j = 1, \dots, J$.

Figure 10.C.5 (right)

The industry marginal cost function.

The derivation of $C'(\cdot)$ just given accords fully with our discussion in Section 5.E. We saw there that the aggregate supply of the J firms, $q(p)$, maximizes aggregate profits given p ; therefore, we can relate $q(\cdot)$ to the industry marginal cost function $C'(\cdot)$ in exactly the same manner as we did in Section 5.D for the case of a single firm's marginal cost function and supply behavior. With convex technologies, the aggregate supply locus for good ℓ therefore coincides with the graph of the industry marginal cost function $C'(\cdot)$, and so $q^{-1}(\cdot) = C'(\cdot)$.¹²

Likewise, at any given level of aggregate demand $\bar{x}$, the *inverse demand function* $P(\bar{x}) = x^{-1}(\bar{x})$ gives the price that results in aggregate demand of $\bar{x}$. That is, when each consumer optimally chooses his demand for good ℓ at this price, total demand exactly equals $\bar{x}$. Note that at these individual demand levels (assuming that they are positive), each consumer's marginal benefit in terms of the numeraire from an additional unit of good ℓ , $\phi'_i(x_i)$, is exactly equal to $P(\bar{x})$. This is illustrated in Figure

11. Formally, the industry marginal cost function $C'(\cdot)$ is the derivative of the aggregate cost function $C(\cdot)$ that gives the total production cost that would be incurred by a central authority who operates all J firms and seeks to produce any given aggregate level of good ℓ at minimum total cost. (See Exercise 10.C.3.)

12. More formally, by Proposition 5.E.1, aggregate supply behavior can be determined by maximizing profit given the aggregate cost function $C(\cdot)$. This yields first-order condition $p = C'(q(p))$. Hence, $q(\cdot) = C'^{-1}(\cdot)$, or equivalently $q^{-1}(\cdot) = C'(\cdot)$.

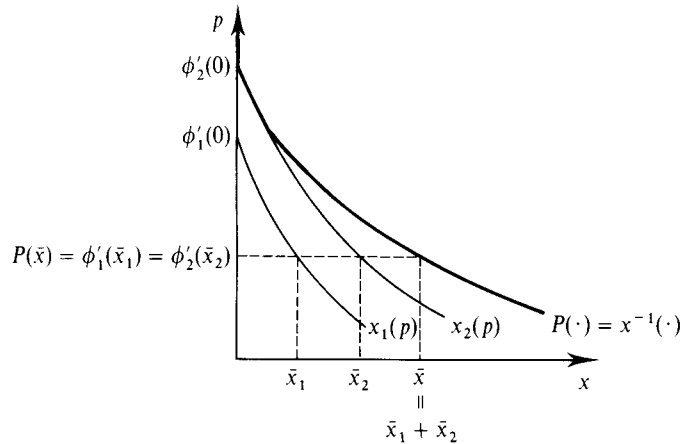


Figure 10.C.6
The inverse demand function.

10.C.6. The value of the inverse demand function at quantity $\bar{x}$, $P(\bar{x})$, can thus be viewed as giving the *marginal social benefit of good ℓ* given that the aggregate quantity $\bar{x}$ is efficiently distributed among the I consumers (see Exercise 10.C.4 for a precise statement of this fact).

Given these interpretations, we can view the competitive equilibrium as involving an aggregate output level at which the marginal social benefit of good ℓ is exactly equal to its marginal cost. This suggests a social optimality property of the competitive allocation, a topic that we investigate further in Section 10.D.

Comparative Statics

It is often of interest to determine how a change in underlying market conditions affects the equilibrium outcome of a competitive market. Such questions may arise, for example, because we may be interested in comparing market outcomes across several similar markets that differ in some measurable way (e.g., we might compare the price of ice cream in a number of cities whose average temperatures differ) or because we want to know how a change in market conditions will alter the outcome in a particular market. The analysis of these sorts of questions is known as *comparative statics analysis*.

As a general matter, we might imagine that each consumer's preferences are affected by a vector of exogenous parameters $\alpha \in \mathbb{R}^M$, so that the utility function $\phi_i(\cdot)$ can be written as $\phi_i(x_i, \alpha)$. Similarly, each firm's technology may be affected by a vector of exogenous parameters $\beta \in \mathbb{R}^S$, so that the cost function $c_j(\cdot)$ can be written as $c_j(q_j, \beta)$. In addition, in some circumstances, consumers and firms face taxes or subsidies that may make the effective (i.e., net of taxes and subsidies) price paid or received differ from the market price p . We let $\hat{p}_i(p, t)$ and $\hat{p}_j(p, t)$ denote, respectively, the effective price paid by consumer i and the effective price received by firm j given tax and subsidy parameters $t \in \mathbb{R}^K$. For example, if consumer i must pay a tax of t_i (in units of the numeraire) per unit of good i purchased, then $\hat{p}_i(p, t) = p + t_i$. If consumer i instead faces a tax that is a percentage t_i of the sales price, then $\hat{p}_i(p, t) = p(1 + t_i)$.

For given values (α, β, t) of the parameters, the $I + J$ equilibrium quantities $(x_1^*, \dots, x_I^*, q_1^*, \dots, q_J^*)$ and the equilibrium price p^* are determined as the solution to the following $I + J + 1$ equations (we assume, for simplicity, that $x_i^* > 0$ for all

i and $q_j^* > 0$ for all j):

$$\phi'_i(x_i^*, \alpha) = \hat{p}_i(p^*, t) \quad i = 1, \dots, I. \quad (10.C.4)$$

$$c'_j(q_j^*, \beta) = \hat{p}_j(p^*, t) \quad j = 1, \dots, J. \quad (10.C.5)$$

$$\sum_{i=1}^I x_i^* = \sum_{j=1}^J q_j^*. \quad (10.C.6)$$

These $I + J + 1$ equations implicitly define the equilibrium allocation and price as functions of the exogenous parameters (α, β, t) . If all the relevant functions are differentiable, we can use the implicit function theorem to derive the marginal change in the equilibrium allocation and price in response to a differential change in the values of these parameters (see Section M.E of the Mathematical Appendix). In Example 10.C.1, we consider one such comparative statics exercise; it is only one among a large number of possibilities that arise naturally in economic applications. (The exercises at the end of this chapter include additional examples.)

Example 10.C.1: Comparative Statics Effects of a Sales Tax. Suppose that a new sales tax is proposed under which consumers must pay an amount $t \geq 0$ (in units of the numeraire) for each unit of good ℓ consumed. We wish to determine the effect of this tax on the market price. Let $x(p)$ and $q(p)$ denote the aggregate demand and supply functions, respectively, for good ℓ in the absence of the tax (we maintain all our previous assumptions regarding these functions).

In terms of our previous notation, the $\phi_i(\cdot)$ and $c_j(\cdot)$ functions do not depend on any exogenous parameters, $\hat{p}_i(p, t) = p + t$ for all i , and $\hat{p}_j(p, t) = p$ for all j . In principle, by substituting these expressions into the system of equilibrium equations (10.C.4) to (10.C.6), we can derive the effect of a marginal increase in the tax on the price by direct use of the implicit function theorem (see Exercise 10.C.5). Here, however, we pursue a more instructive way to get the answer. In particular, note that aggregate demand with a tax of t and price p is exactly $x(p + t)$ because the tax is equivalent for consumers to the price being increased by t . Thus, the equilibrium market price when the tax is t , which we denote by $p^*(t)$, must satisfy

$$x(p^*(t) + t) = q(p^*(t)). \quad (10.C.7)$$

Suppose that we now want to determine the effect on prices paid and received of a marginal increase in the tax. Assuming that $x(\cdot)$ and $q(\cdot)$ are differentiable at $p = p^*(t)$, differentiating (10.C.7) yields

$$p^{*'}(t) = -\frac{x'(p^*(t) + t)}{x'(p^*(t) + t) - q'(p^*(t))}. \quad (10.C.8)$$

It is immediate from (10.C.8) and our assumptions on $x'(\cdot)$ and $q'(\cdot)$ that $-1 \leq p^{*'}(t) < 0$ at any t . Therefore, the price $p^*(t)$ received by producers falls as t increases while the overall cost of the good to consumers $p^*(t) + t$ rises (weakly). The total quantities produced and consumed fall (again weakly). See Figure 10.C.7(a), where the equilibrium level of aggregate consumption at tax rate t is denoted by $x^*(t)$. Notice from (10.C.8) that when $q'(p^*(t))$ is large we have $p^{*'}(t) \approx 0$, and so the price received by the firms is hardly affected by the tax; nearly all the impact of the tax is felt by consumers. In contrast, when $q'(p^*(t)) = 0$, we have $p^{*'}(t) = -1$, and so the impact of the tax is felt entirely by the firms. Figures 10.C.7(b) and (c) depict these two cases.

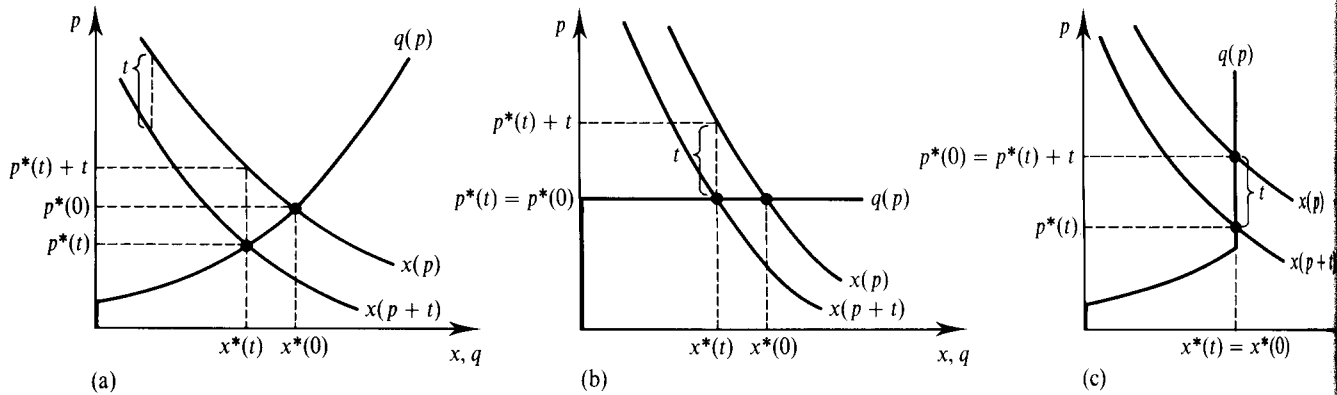


Figure 10.C.7 Comparative statics effects of a sales tax.

By substituting into (10.C.8) for $x'(\cdot)$ and $q'(\cdot)$, the marginal change in p^* can be expressed in terms of derivatives of the underlying individual utility and cost functions. For example, if we let $p^* = p^*(0)$ be the pretax price, we see that

$$p^{*'}(0) = - \frac{\sum_{i=1}^I [\phi_i''(x_i(p^*))]^{-1}}{\sum_{i=1}^I [\phi_i''(x_i(p^*))]^{-1} - \sum_{j=1}^J [c_j''(q_j(p^*))]^{-1}}.$$

We have assumed throughout this section that consumers' preferences and firms' technologies are convex (and strictly so in the case of consumer preferences). What if this is not the case? Figure 10.C.8 illustrates one problem that can then arise; it shows the demand function and

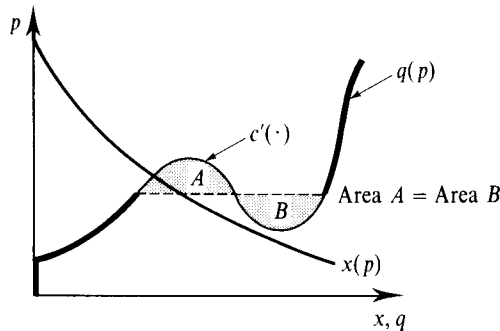


Figure 10.C.8 Nonexistence of competitive equilibrium with a nonconvex technology.

supply correspondence for an economy in which there is a single firm (so $J = 1$).¹³ This firm's cost function $c(\cdot)$ is continuous and differentiable but not convex. In the figure, the light curve is the graph of the firm's marginal cost function $c'(\cdot)$. As the figure illustrates, $c'(\cdot)$ fails to be nondecreasing. The heavier curve is the firm's actual supply correspondence $q(\cdot)$ (you should verify that it is determined as indicated in the figure).¹⁴ The graph of the supply correspondence no longer coincides with the marginal cost curve and, as is evident in the figure, no intersection exists between the graph of the supply correspondence and the demand curve. Thus, in this case, *no competitive equilibrium exists*.

13. We set $J = 1$ here solely for expositional purposes.

14. See Section 5.D for a more detailed discussion of the relation between a firm's supply correspondence and its marginal cost function when its technology is nonconvex.

This observation suggests that convexity assumptions are key to the existence of a competitive equilibrium. We shall confirm this in Chapter 17, where we provide a more general discussion of the conditions under which existence of a competitive equilibrium is assured.

10.D The Fundamental Welfare Theorems in a Partial Equilibrium Context

In this section, we study the properties of Pareto optimal allocations in the framework of the two-good quasilinear economy introduced in Section 10.C, and we establish a fundamental link between the set of Pareto optimal allocations and the set of competitive equilibria.

The identification of Pareto optimal allocations is considerably facilitated by the quasilinear specification. In particular, *when consumer preferences are quasilinear, the boundary of the economy's utility possibility set is linear* (see Section 10.B for the definition of this set) *and all points in this boundary are associated with consumption allocations that differ only in the distribution of the numeraire among consumers.*

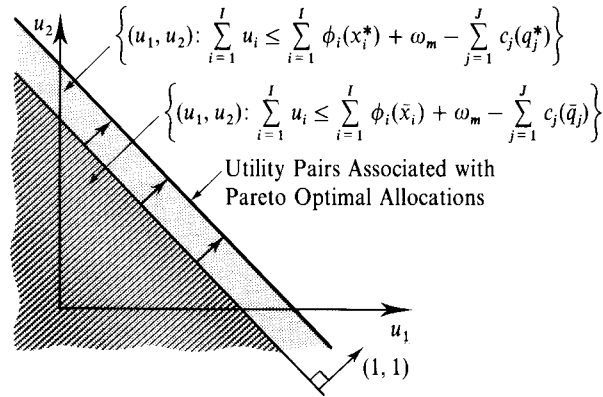
To see this important fact, suppose that we fix the consumption and production levels of good ℓ at $(\bar{x}_1, \dots, \bar{x}_I, \bar{q}_1, \dots, \bar{q}_J)$. With these production levels, the total amount of the numeraire available for distribution among consumers is $\omega_m - \sum_j c_j(\bar{q}_j)$. Because the quasilinear form of the utility functions allows for an unlimited unit-for-unit transfer of utility across consumers through transfers of the numeraire, the set of utilities that can be attained for the I consumers by appropriately distributing the available amounts of the numeraire is given by

$$\left\{ (u_1, \dots, u_I) : \sum_{i=1}^I u_i \leq \sum_{i=1}^I \phi_i(\bar{x}_i) + \omega_m - \sum_{j=1}^J c_j(\bar{q}_j) \right\}. \quad (10.D.1)$$

The boundary of this set is a hyperplane with normal vector $(1, \dots, 1)$. The set is depicted for the case $I = 2$ by the hatched set in Figure 10.D.1.

Note that by altering the consumption and production levels of good ℓ , we necessarily shift the boundary of this set in a parallel manner. Thus, every Pareto optimal allocation must involve the quantities $(x_1^*, \dots, x_I^*, q_1^*, \dots, q_J^*)$ that extend this boundary as far out as possible, as illustrated by the heavily drawn boundary of the shaded utility possibility set in Figure 10.D.1. We call these quantities the *optimal consumption and production levels* for good ℓ . As long as these optimal consumption and production levels for good ℓ are uniquely determined, Pareto optimal allocations can differ only in the distribution of the numeraire among consumers.¹⁵

15. The optimal individual production levels need not be unique if firms' cost functions are convex but not strictly so. Indeterminacy of optimal individual production levels arises, for example, when all firms have identical constant returns to scale technologies. However, under our assumptions that the $\phi_i(\cdot)$ functions are strictly concave and that the $c_j(\cdot)$ functions are convex, the optimal individual consumption levels of good ℓ are necessarily unique and, hence, so is the optimal aggregate production level $\sum_j q_j^*$ of good ℓ . This implies that, under our assumptions, the consumption allocations in two different Pareto optimal allocations can differ only in the distribution of numeraire among consumers. If, moreover, the $c_j(\cdot)$ functions are strictly convex, then the optimal individual production levels are also uniquely determined. (See Exercise 10.D.1.)

**Figure 10.D.1**

The utility possibility set in a quasilinear economy.

It follows from expression (10.D.1) that the optimal consumption and production levels of good ℓ can be obtained as the solution to

$$\begin{aligned} \text{Max}_{\substack{(x_1, \dots, x_I) \geq 0 \\ (q_1, \dots, q_J) \geq 0}} \quad & \sum_{i=1}^I \phi_i(x_i) - \sum_{j=1}^J c_j(q_j) + \omega_m \quad (10.D.2) \\ \text{s.t.} \quad & \sum_{i=1}^I x_i - \sum_{j=1}^J q_j = 0. \end{aligned}$$

The value of the term $\sum_i \phi_i(x_i) - \sum_j c_j(q_j)$ in the objective function of problem (10.D.2) is known as the *Marshallian aggregate surplus* (or, simply, the *aggregate surplus*). It can be thought of as the total utility generated from consumption of good ℓ less its costs of production (in terms of the numeraire). The optimal consumption and production levels for good ℓ maximize this aggregate surplus measure.

Given our convexity assumptions, the first-order conditions of problem (10.D.2) yield necessary and sufficient conditions that characterize the optimal quantities. If we let μ be the multiplier on the constraint in problem (10.D.2), the $I + J$ optimal values $(x_1^*, \dots, x_I^*, q_1^*, \dots, q_J^*)$ and the multiplier μ satisfy the following $I + J + 1$ conditions:

$$\mu \leq c'_j(q_j^*), \quad \text{with equality if } q_j^* > 0 \quad j = 1, \dots, J. \quad (10.D.3)$$

$$\phi'_i(x_i^*) \leq \mu, \quad \text{with equality if } x_i^* > 0 \quad i = 1, \dots, I. \quad (10.D.4)$$

$$\sum_{i=1}^I x_i^* = \sum_{j=1}^J q_j^*. \quad (10.D.5)$$

These conditions should look familiar: They exactly parallel conditions (10.C.1) to (10.C.3) in Section 10.C, with μ replacing p^* . This observation has an important implication. We can immediately infer from it that any competitive equilibrium outcome in this model is Pareto optimal because any competitive equilibrium allocation has consumption and production levels of good ℓ , $(x_1^*, \dots, x_I^*, q_1^*, \dots, q_J^*)$, that satisfy conditions (10.D.3) to (10.D.5) when we set $\mu = p^*$. Thus, we have established the *first fundamental theorem of welfare economics* (Proposition 10.D.1) in the context of this quasilinear two-good model.

Proposition 10.D.1: (*The First Fundamental Theorem of Welfare Economics*) If the price p^* and allocation $(x_1^*, \dots, x_I^*, q_1^*, \dots, q_J^*)$ constitute a competitive equilibrium, then this allocation is Pareto optimal.

The first fundamental welfare theorem establishes conditions under which market equilibria are necessarily Pareto optimal. It is a formal expression of Adam Smith's "invisible hand" and is a result that holds with considerable generality (see Section 16.C for a much more extensive discussion). Equally important, however, are the conditions under which it fails to hold. In the models for which we establish the first fundamental welfare theorem here and in Section 16.C, markets are "complete" in the sense that there is a market for every relevant commodity and all market participants act as price takers. In Chapters 11 to 14, we study situations in which at least one of these conditions fails, and market outcomes fail to be Pareto optimal as a result.

We can also develop a converse to Proposition 10.D.1, known as the *second fundamental theorem of welfare economics*. In Section 10.C, we saw that good ℓ 's equilibrium price p^* , its equilibrium consumption and production levels $(x_1^*, \dots, x_I^*, q_1^*, \dots, q_J^*)$, and firms' profits are unaffected by changes in consumers' wealth levels. As a result, a transfer of one unit of the numeraire from consumer i to consumer i' will cause each of these consumers' equilibrium consumption of the numeraire to change by exactly the amount of the transfer and will cause no other changes. Thus, by appropriately transferring endowments of the numeraire commodity, the resulting competitive equilibrium allocation can be made to yield any utility vector in the boundary of the utility possibility set. The second welfare theorem therefore tells us that, in this two-good quasilinear economy, a central authority interested in achieving a particular Pareto optimal allocation can always implement this outcome by transferring the numeraire among consumers and then "allowing the market to work." This is stated formally in Proposition 10.D.2.

Proposition 10.D.2: (*The Second Fundamental Theorem of Welfare Economics*) For any Pareto optimal levels of utility $(u_1^*, \dots, u_I^*)$, there are transfers of the numeraire commodity $(T_1, \dots, T_I)$ satisfying $\sum_i T_i = 0$, such that a competitive equilibrium reached from the endowments $(\omega_{m1} + T_1, \dots, \omega_{mI} + T_I)$ yields precisely the utilities $(u_1^*, \dots, u_I^*)$.

In Section 16.D, we study the conditions under which the second welfare theorem holds in more general competitive economies. A critical requirement, in addition to those needed for the first welfare theorem, turns out to be convexity of preferences and production sets, an assumption we have made in the model under consideration here. In contrast, we shall see in Chapter 16 that no such convexity assumptions are needed for the first welfare theorem.

The correspondence between p and μ in the equilibrium conditions (10.C.1) to (10.C.3) and the Pareto optimality conditions (10.D.3) to (10.D.5) is worthy of emphasis: The competitive price is exactly equal to the shadow price on the resource constraint for good ℓ in the Pareto optimality problem (10.D.2). In this sense, then, we can say that a good's price in a competitive equilibrium reflects precisely its marginal social value. In a competitive equilibrium, each firm, by operating at a point where price equals marginal cost, equates its marginal production cost to the marginal social value of its output. Similarly, each consumer, by consuming up to the point where marginal utility from a good equals its price, is at a point where the marginal benefit from consumption of the good exactly equals its marginal cost. This correspondence between equilibrium market prices and optimal shadow prices holds

quite generally in competitive economies (see Section 16.F for further discussion of this point).

An alternative way to characterize the set of Pareto optimal allocations is to solve

$$\begin{aligned}
 & \text{Max}_{\{x_i, m_i\}_{i=1}^I, \{z_j, q_j\}_{j=1}^J} m_1 + \phi_1(x_1) & (10.D.6) \\
 & \text{s.t.} \quad (1) \quad m_i + \phi_i(x_i) \geq \bar{u}_i \quad i = 2, \dots, I \\
 & \quad (2\ell) \quad \sum_{i=1}^I x_i - \sum_{j=1}^J q_j \leq 0 \\
 & \quad (2m) \quad \sum_{i=1}^I m_i + \sum_{j=1}^J z_j \leq \omega_m \\
 & \quad (3) \quad z_j \geq c_j(q_j) \quad j = 1, \dots, J.
 \end{aligned}$$

Problem (10.D.6) expresses the Pareto optimality problem as one of trying to maximize the well-being of individual 1 subject to meeting certain required utility levels for the other individuals in the economy [constraints (1)], resource constraints [constraints (2 ℓ) and (2m)], and technological constraints [constraints (3)]. By solving problem (10.D.6) for various required levels of utility for these other individuals, $(\bar{u}_2, \dots, \bar{u}_I)$, we can identify all the Pareto optimal outcomes for this economy (see Exercise 10.D.3; more generally, we can do this whenever consumer preferences are strongly monotone). Exercise 10.D.4 asks you to derive conditions (10.D.3) to (10.D.5) in this alternative manner.

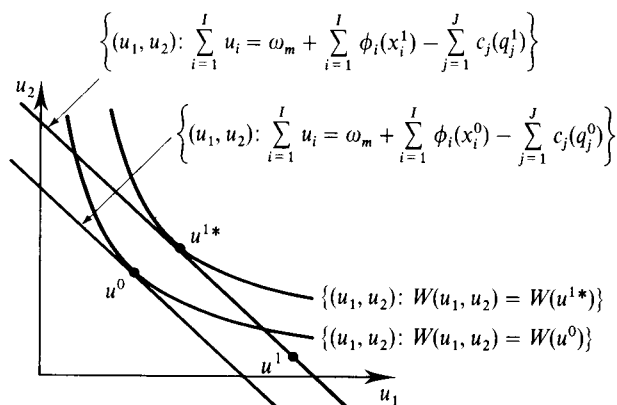
10.E Welfare Analysis in the Partial Equilibrium Model

It is often of interest to measure the change in the level of social welfare that would be generated by a change in market conditions such as an improvement in technology, a new government tax policy, or the elimination of some existing market imperfection. In the partial equilibrium model, it is particularly simple to carry out this welfare analysis. This fact accounts to a large extent for the popularity of the model.

In the discussion that follows, we assume that the welfare judgments of society are embodied in a social welfare function $W(u_1, \dots, u_I)$ assigning a social welfare value to every utility vector $(u_1, \dots, u_I)$ (see Chapters 4, 16, and 22 for more on this concept). In addition, we suppose that (as in the theory of the normative representative consumer discussed in Section 4.D) there is some central authority who redistributes wealth by means of transfers of the numeraire commodity in order to maximize social welfare.¹⁶ The critical simplification offered by the quasilinear specification of individual utility functions is that when there is a central authority who redistributes wealth in this manner, *changes in social welfare can be measured by changes in the Marshallian aggregate surplus* (introduced in Section 10.D) *for any social welfare function that society may have*.

To see this point (which we have in fact already examined in Example 4.D.2), consider some given consumption and production levels of good ℓ , $(x_1, \dots, x_I, q_1, \dots, q_J)$,

16. As in Section 4.D, we assume that consumers treat these transfers as independent of their own actions; that is, in the standard terminology, they are *lump-sum* transfers. You should think of the central authority as making the transfers prior to the opening of markets.

**Figure 10.E.1**

With lump-sum redistribution occurring to maximize social welfare, changes in welfare correspond to changes in aggregate surplus in a quasilinear model.

having $\sum_i x_i = \sum_j q_j$. From Section 10.D and Figure 10.D.1 we know that the utility vectors $(u_1, \dots, u_I)$ that are achievable through reallocation of the numeraire given these consumption and production levels of good ℓ are

$$\left\{ (u_1, \dots, u_I) : \sum_{i=1}^I u_i \leq \omega_m + \sum_{i=1}^I \phi_i(x_i) - \sum_{j=1}^J c_j(q_j) \right\}.$$

Now, if a central authority is redistributing the numeraire to maximize $W(u_1, \dots, u_I)$, the ultimate maximized value of welfare must be greater the larger this set is (i.e., the farther out the boundary of the set is). Hence, we see that a change in the consumption and production levels of good ℓ leads to an increase in welfare (given optimal redistribution of the numeraire) if and only if it increases the Marshallian aggregate surplus

$$S(x_1, \dots, x_I, q_1, \dots, q_J) = \sum_{i=1}^I \phi_i(x_i) - \sum_{j=1}^J c_j(q_j). \quad (10.E.1)$$

Figure 10.E.1 provides an illustration. It shows three utility vectors for the case $I = 2$: An initial utility vector $u^0 = (u_1^0, u_2^0)$ associated with an allocation in which the consumption and production levels of good ℓ are $(x_1^0, \dots, x_I^0, q_1^0, \dots, q_J^0)$ and in which the wealth distribution has been optimized, a utility vector $u^1 = (u_1^1, u_2^1)$ that results from a change in the consumption and production levels of good ℓ to $(x_1^1, \dots, x_I^1, q_1^1, \dots, q_J^1)$ in the absence of any transfers of the numeraire, and a utility vector $u^{1*} = (u_1^{1*}, u_2^{1*})$ that results from this change once redistribution of the numeraire occurs to optimize social welfare. As can be seen in the figure, the change increases aggregate surplus and also increases welfare once optimal transfers of the numeraire occur, even though welfare would decrease in the absence of the transfers. Thus, as long as redistribution of wealth is occurring to maximize a social welfare function, changes in welfare can be measured by changes in Marshallian aggregate surplus (to repeat: for *any* social welfare function).¹⁷

In many circumstances of interest, the Marshallian surplus has a convenient and

17. Notice that no transfers would be necessary in the special case in which the social welfare function is in fact the “utilitarian” social welfare function $\sum_i u_i$; in this case, it is sufficient that all available units of the numeraire go to consumers (i.e., none goes to waste or is otherwise withheld).

historically important formulation in terms of areas lying vertically between the aggregate demand and supply functions for good ℓ .

To expand on this point, we begin by making two key assumptions. Denoting by $x = \sum_i x_i$ the aggregate consumption of good ℓ , we assume, first, that for any x , the individual consumptions of good ℓ are distributed optimally across consumers. That is, recalling our discussion of the inverse demand function $P(\cdot)$ in Section 10.C (see Figure 10.C.6), that we have $\phi'_i(x_i) = P(x)$ for every i . This condition will be satisfied if, for example, consumers act as price-takers and all consumers face the same price. Similarly, denoting by $q = \sum_j q_j$ the aggregate output of good ℓ , we assume that the production of any total amount q is distributed optimally across firms. That is, recalling our discussion of the industry marginal cost curve $C'(\cdot)$ in Section 10.C (see Figure 10.C.5), that we have $c'_j(q_j) = C'(q)$ for every j . This will be satisfied if, for example, firms act as price takers and all firms face the same price. Observe that we do not require that the price faced by consumers and firms be the same.¹⁸

Consider now a differential change $(dx_1, \dots, dx_I, dq_1, \dots, dq_J)$ in the quantities of good ℓ consumed and produced satisfying $\sum_i dx_i = \sum_j dq_j$, and denote $dx = \sum_i dx_i$. The change in aggregate Marshallian surplus is then

$$dS = \sum_{i=1}^I \phi'_i(x_i) dx_i - \sum_{j=1}^J c'_j(q_j) dq_j. \quad (10.E.2)$$

Since $\phi'_i(x_i) = P(x)$ for all i , and $c'_j(q_j) = C'(q)$ for all j , we get

$$dS = P(x) \sum_{i=1}^I dx_i - C'(q) \sum_{j=1}^J dq_j. \quad (10.E.3)$$

Finally, since $x = q$ (by market feasibility) and $\sum_j dq_j = \sum_i dx_i = dx$, this becomes

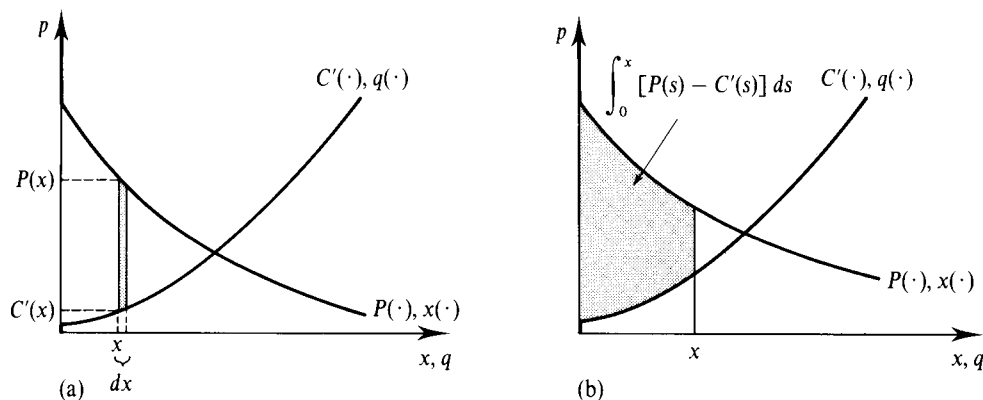
$$dS = [P(x) - C'(x)] dx. \quad (10.E.4)$$

This differential change in Marshallian surplus is depicted in Figure 10.E.2(a). Expression (10.E.4) is quite intuitive; it tells us that starting at aggregate consumption level x the marginal effect on social welfare of an increase in the aggregate quantity consumed, dx , is equal to consumers' marginal benefit from this consumption, $P(x) dx$, less the marginal cost of this extra production, $C'(x) dx$ (both in terms of the numeraire).

We can also integrate (10.E.4) to express the total value of the aggregate Marshallian surplus at the aggregate consumption level x , denoted $S(x)$, in terms of an integral of the difference between the inverse demand function and the industry marginal cost function,

$$S(x) = S_0 + \int_0^x [P(s) - C'(s)] ds, \quad (10.E.5)$$

18. For example, consumers may face a tax per unit purchased that makes the price they pay differ from the price received by the firms (see Example 10.C.1). The assumptions made here also hold in the monopoly model to be studied in Section 12.B. In that model, there is a single firm (and so there is no issue of optimal allocation of production), and all consumers act as price takers facing the same price. An example where the assumption of an optimal allocation of production is not valid is the Cournot duopoly model of Chapter 12 when firms have different efficiencies. There, firms with different costs have different levels of marginal cost in an equilibrium.

**Figure 10.E.2**

(a) A differential change in Marshallian surplus. (b) The Marshallian surplus at aggregate consumption level x .

where S_0 is a constant of integration equal to the value of the aggregate surplus when there is no consumption or production of good ℓ [it is equal to zero if $c_j(0) = 0$ for all j]. The integral in (10.E.5) is depicted in Figure 10.E.2(b); it is exactly equal to the area lying vertically between the aggregate demand and supply curves for good ℓ up to quantity x .

Note from (10.E.5) that the value of the aggregate Marshallian surplus is maximized at the aggregate consumption level x^* such that $P(x^*) = C'(x^*)$, which is exactly the competitive equilibrium aggregate consumption level.¹⁹ This accords with Proposition 10.D.1, the first fundamental welfare theorem, which states that the competitive allocation is Pareto optimal.

Example 10.E.1: The Welfare Effects of a Distortionary Tax. Consider again the commodity tax problem studied in Example 10.C.1. Suppose now that the welfare authority keeps a balanced budget and returns the tax revenue raised to consumers by means of lump-sum transfers. What impact does this tax-and-transfer scheme have on welfare?²⁰

To answer this question, it is convenient to let $(x_1^*(t), \dots, x_I^*(t), q_1^*(t), \dots, q_J^*(t))$ and $p^*(t)$ denote the equilibrium consumption, production, and price levels of good ℓ when the tax rate is t . Note that $\phi_i'(x_i^*(t)) = p^*(t) + t$ for all i and that $c_j'(q_j^*(t)) = p^*(t)$ for all j . Thus, letting $x^*(t) = \sum_i x_i^*(t)$ and $S^*(t) = S(x^*(t))$, we can use (10.E.5) to express the change in aggregate Marshallian surplus resulting from

19. To see this, check first that $S''(x) \leq 0$ at all x . Hence, $S(\cdot)$ is a concave function and therefore $x^* > 0$ maximizes aggregate surplus if and only if $S'(x^*) = 0$. Then verify that $S'(x) = P(x) - C'(x)$ at all $x > 0$.

20. This problem is closely related to that studied in Example 3.I.1 (we could equally well motivate the analysis here by asking, as we did there, about the welfare cost of the distortionary tax relative to the use of a lump-sum tax that raises the same revenue; the measure of deadweight loss that emerges would be the same as that developed here). The discussion that follows amounts to an extension, in the quasilinear context, of the analysis of Example 3.I.1 to situations with many consumers and the presence of firms. For an approach that uses the theory of a normative representative consumer presented in Section 4.D, see the small-type discussion at the end of this section.

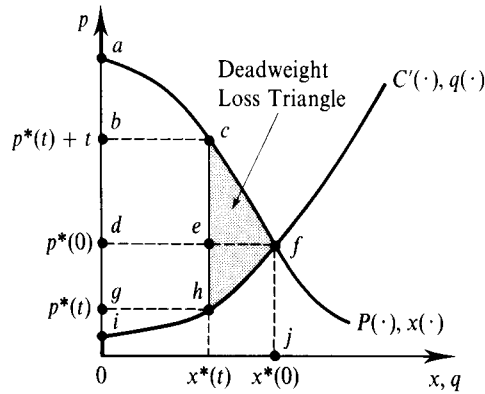


Figure 10.E.3
The deadweight welfare loss from distortional taxation.

the introduction of the tax as

$$S^*(t) - S^*(0) = \int_{x^*(0)}^{x^*(t)} [P(s) - C'(s)] ds. \quad (10.E.6)$$

Expression (10.E.6) is negative because $x^*(t) < x^*(0)$ (recall the analysis of Example 10.C.1) and $P(x) \geq C'(x)$ for all $x \leq x^*(0)$, with strict inequality for $x < x^*(0)$. Hence, social welfare is optimized by setting $t = 0$. The loss in welfare from $t > 0$ is known as the *deadweight loss of distortional taxation* and is equal to the area of the shaded region in Figure 10.E.3, called the *deadweight loss triangle*.

Notice that since $S^*(t) = [P(x^*(t)) - C'(x^*(t))]x^*(t)$, we have $S^*(0) = 0$. That is, starting from a position without any tax, the first-order welfare effect of an infinitesimal tax is zero. Only as the tax rate increases above zero does the marginal effect become strictly negative. This is as it should be: if we start at an (interior) welfare maximum, then a small displacement from the optimum cannot have a first-order effect on welfare.

It is sometimes of interest to distinguish between the various components of aggregate Marshallian surplus that accrue directly to consumers, firms, and the tax authority.²¹ The *aggregate consumer surplus* when consumers' effective price is $\hat{p}$ and therefore aggregate consumption is $x(\hat{p})$ is defined as the gross consumer benefits from consumption of good ℓ minus the consumers' total expenditure on this good (the latter is the cost to consumers in terms of forgone consumption of the numeraire):

$$CS(\hat{p}) = \sum_{i=1}^I \phi_i(x_i(\hat{p})) - \hat{p}x(\hat{p}).$$

Using again the fact that consumption is distributed optimally, we have

$$\begin{aligned} CS(\hat{p}) &= \int_0^{x(\hat{p})} P(s) ds - \hat{p}x(\hat{p}) \\ &= \int_0^{x(\hat{p})} [P(s) - \hat{p}] ds. \end{aligned} \quad (10.E.7)$$

21. For example, if the set of active consumers of good ℓ is distinct from the set of owners of the firms producing the good, then this distinction tells us something about the distributional effects of the tax in the absence of transfers between owners and consumers.

Finally, the integral in (10.E.7) is equal to²²

$$CS(\hat{p}) = \int_{\hat{p}}^{\infty} x(s) ds. \quad (10.E.8)$$

Thus, because consumers face an effective price of $p^*(t) + t$ when the tax is t , the change in consumer surplus from imposition of the tax is

$$CS(p^*(t) + t) - CS(p^*(0)) = - \int_{p^*(0)}^{p^*(t)+t} x(s) ds. \quad (10.E.9)$$

In Figure 10.E.3, the reduction in consumer surplus is depicted by area $(dbcf)$.

The aggregate profit, or *aggregate producer surplus*, when firms face effective price $\hat{p}$ is

$$\Pi(\hat{p}) = \hat{p}q(\hat{p}) - \sum_{j=1}^J c_j(q_j(\hat{p})).$$

Again, using the optimality of the allocation of production across firms, we have²³

$$\Pi(\hat{p}) = \Pi_0 + \int_0^{q(\hat{p})} [\hat{p} - C'(s)] ds \quad (10.E.10)$$

$$= \Pi_0 + \int_0^{\hat{p}} q(s) ds, \quad (10.E.11)$$

where Π_0 is a constant of integration equal to profits when $q_j = 0$ for all j [$\Pi_0 = 0$ if $c_j(0) = 0$ for all j]. Since producers pay no tax, they face price $p^*(t)$ when the tax rate is t . The change in producer surplus is therefore

$$\Pi(p^*(t)) - \Pi(p^*(0)) = - \int_{p^*(t)}^{p^*(0)} q(s) ds. \quad (10.E.12)$$

The reduction in producer surplus is depicted by area $(gdfh)$ in Figure 10.E.3.

Finally, the *tax revenue* is $tx^*(t)$; it is depicted in Figure 10.E.3 by area $(gbch)$.

The total deadweight welfare loss from the tax is then equal to the sum of the reductions in consumer and producer surplus less the tax revenue. ■

The welfare measure developed here is closely related to our discussion of normative representative consumers in Section 4.D. We showed there that if a central authority is redistributing wealth to maximize a social welfare function given prices p , leading to a wealth distribution rule $(w_1(p, w), \dots, w_I(p, w))$, then there is a normative representative consumer with indirect utility function $v(p, w)$ whose demand $x(p, w)$ is exactly equal to aggregate demand [i.e., $x(p, w) = \sum_i x_i(p, w_i(p, w))$] and whose utility can be used as a measure of social welfare. Recalling our discussion in Section 3.I, this means that we can measure the change in welfare resulting from a price-wealth change by adding the representative consumer's

22. This can be seen geometrically. For example, when $\hat{p} = p^*(0)$, the integrals in both (10.E.7) and (10.E.8) are equal to area (daf) in Figure 10.E.3. Formally, the equivalence follows from a change of variables and integration by parts (see Exercise 10.E.2).

23. When $\hat{p} = p^*(0)$, the integrals in both (10.E.10) and (10.E.11) are equal to area (idf) in Figure 10.E.3. The equivalence of these two integrals again follows formally by a change of variables and integration by parts.

compensating or equivalent variation for the price change to the change in the representative consumer's wealth (see Exercise 3.I.12). But in the quasilinear case, the representative consumer's compensating and equivalent variations are the same and can be calculated by direct integration of the representative consumer's Walrasian demand function, that is, by integration of the aggregate demand function. Hence, in Example 10.E.1, the representative consumer's compensating variation for the price change is exactly equal to the change in aggregate consumer surplus, expression (10.E.9). The change in the representative consumer's wealth, on the other hand, is equal to the change in aggregate profits plus the tax revenue rebated to consumers. Thus, the total welfare change arising from the introduction of the tax-and-transfer scheme, as measured using the normative representative consumer, is exactly equal to the deadweight loss calculated in Example 10.E.1.²⁴

Another way to justify the use of aggregate surplus as a welfare measure in the quasilinear model is as a measure of *potential Pareto improvement*. Consider the tax example. We could say that a change in the tax represents a *potential* Pareto improvement if there is a set of lump-sum transfers of the numeraire that would make all consumers better off than they were before the tax change. In the present quasilinear context, this is true if and only if aggregate surplus increases with the change in the tax. This approach is sometimes referred to as the *compensation principle* because it asks whether, in principle, it is possible given the change for the winners to compensate the losers so that all are better off than before. (See also the discussion in Example 4.D.2 and especially Section 22.C.)

We conclude this section with a warning: When the numeraire represents many goods, the welfare analysis we have performed is justified only if the prices of goods other than good ℓ are undistorted in the sense that they equal these goods' true marginal utilities and production costs. Hence, these other markets must be competitive, and all market participants must face the same price. If this condition does not hold, then the costs of production faced by producers of good ℓ do not reflect the true social costs incurred from their use of these goods as inputs. Exercise 10.G.3 provides an illustration of this problem.

10.F Free-Entry and Long-Run Competitive Equilibria

Up to this point, we have taken the set of firms and their technological capabilities as fixed. In this section, we consider the case in which an infinite number of firms can potentially be formed, each with access to the most efficient production technology. Moreover, firms may enter or exit the market in response to profit opportunities. This scenario, known as a situation of *free entry*, is often a reasonable approximation when we think of long-run outcomes in a market. In the discussion that follows, we introduce and study a notion of *long-run competitive equilibrium* and then discuss how this concept can be used to analyze long-run and short-run comparative statics effects.

To begin, suppose that each of an infinite number of potential firms has access to a technology for producing good ℓ with cost function $c(q)$, where q is the *individual* firm's output of good ℓ . We assume that $c(0) = 0$; that is, a firm can earn zero profits by simply deciding to be inactive and setting $q = 0$. In the terminology of Section

24. This deadweight loss measure corresponds also to the measure developed for the one-consumer case in Example 3.I.1, where we implicitly limited ourselves to the case in which the taxed good has a constant unit cost.

5.B, there are no sunk costs in the long run. The aggregate demand function is $x(\cdot)$, with inverse demand function $P(\cdot)$.

In a long-run competitive equilibrium, we would like to determine not only the price and output levels for the firms but also the number of firms that are active in the industry. Given our assumption of identical firms, we focus on equilibria in which all active firms produce the same output level, so that a long-run competitive equilibrium can be described by a triple (p, q, J) formed by a price p , an output per firm q , and an integer number of active firms J (hence the total industry output is $Q = Jq$).²⁵ The central assumption determining the number of active firms is one of free entry and exit: A firm will enter the market if it can earn positive profits at the going market price and will exit if it can make only negative profits at any positive production level given this price. If all firms, active and potential, take prices as unaffected by their own actions, this implies that active firms must earn exactly zero profits in any long-run competitive equilibrium; otherwise, we would have either no firms willing to be active in the market (if profits were negative) or an infinite number of firms entering the market (if profits were positive). This leads us to the formulation given in Definition 10.F.1.

Definition 10.F.1: Given an aggregate demand function $x(p)$ and a cost function $c(q)$ for each potentially active firm having $c(0) = 0$, a triple (p^*, q^*, J^*) is a *long-run competitive equilibrium* if

- (i) q^* solves $\text{Max}_{q \geq 0} p^*q - c(q)$ (Profit maximization)
- (ii) $x(p^*) = J^*q^*$ (Demand = supply)
- (iii) $p^*q^* - c(q^*) = 0$ (Free Entry Condition).

The long-run equilibrium price can be thought of as equating demand with long-run supply, where the long-run supply takes into account firms' entry and exit decisions. In particular, if $q(\cdot)$ is the supply correspondence of an individual firm with cost function $c(\cdot)$ and $\pi(\cdot)$ is its profit function, we can define a *long-run aggregate supply correspondence* by²⁶

$$Q(p) = \begin{cases} \infty & \text{if } \pi(p) > 0, \\ \{Q \geq 0: Q = Jq \text{ for some integer } J \geq 0 \text{ and } q \in q(p)\} & \text{if } \pi(p) = 0. \end{cases}$$

If $\pi(p) > 0$, then every firm wants to supply an amount strictly bounded away from zero. Hence, the aggregate supply is infinite. If $\pi(p) = 0$ and $Q = Jq$ for some $q \in q(p)$, then we can have J firms each supply q and have the rest remain inactive [since $c(0) = 0$, this is a profit-maximizing choice for the inactive firms as well]. With this

25. The assumption that all active firms produce the same output level is without loss of generality whenever $c(\cdot)$ is strictly convex on the set $(0, \infty]$. A firm's supply correspondence can then include at most one positive output level at any given price p .

26. In terms of the basic properties of production sets presented in Section 5.B, the long-run supply correspondence is the supply correspondence of the production set Y^+ , where Y is the production set associated with the individual firm [i.e., with $c(\cdot)$], and Y^+ is its "additive closure" (i.e., the smallest set that contains Y and is additive: $Y^+ + Y^+ \subset Y^+$; see Exercise 5.B.4).

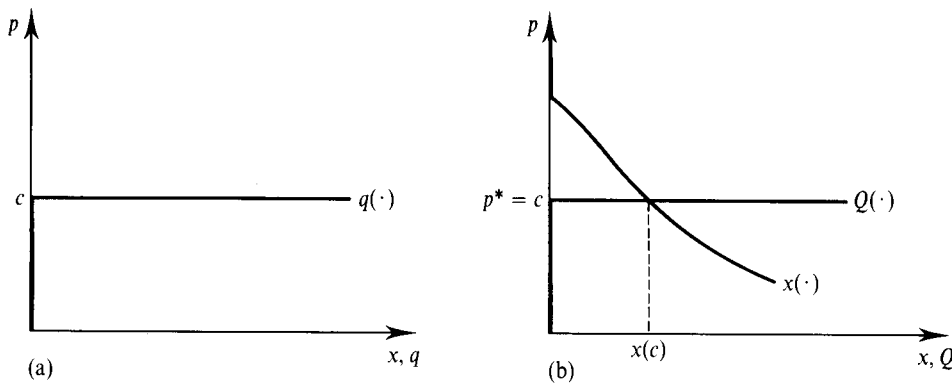


Figure 10.F.1
Long-run competitive equilibrium with constant returns to scale. (a) A firm's supply correspondence. (b) Long-run equilibrium.

notion of a long-run supply correspondence, p^* is a long-run competitive equilibrium price if and only if $x(p^*) \in Q(p^*)$.²⁷

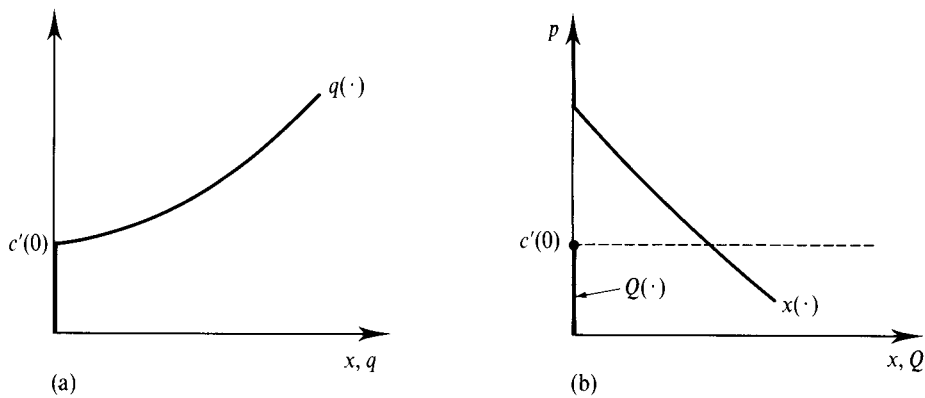
We now investigate this long-run competitive equilibrium notion. Consider first the case in which the cost function $c(\cdot)$ exhibits constant returns to scale, so that $c(q) = cq$ for some $c > 0$, and assume that $x(c) > 0$. In this case, condition (i) of Definition 10.F.1 tells us that in any long-run competitive equilibrium we have $p^* \leq c$ (otherwise, there is no profit-maximizing production). However, at any such price, aggregate consumption is strictly positive since $x(c) > 0$, so condition (ii) requires that $q^* > 0$. By condition (iii), we must have $(p^* - c)q^* = 0$. Hence, we conclude that $p^* = c$ and aggregate consumption is $x(c)$. Note, however, that J^* and q^* are *indeterminate*: any J^* and q^* such that $J^*q^* = x(c)$ satisfies conditions (i) and (ii).

Figure 10.F.1 depicts this long-run equilibrium. The supply correspondence of an individual firm $q(\cdot)$ is illustrated in Figure 10.F.1(a); Figure 10.F.1(b) shows the long-run equilibrium price and aggregate output as the intersection of the graph of the aggregate demand function $x(\cdot)$ with the graph of the long-run aggregate supply correspondence

$$Q(p) = \begin{cases} \infty & \text{if } p > c \\ [0, \infty) & \text{if } p = c \\ 0 & \text{if } p < c. \end{cases}$$

We move next to the case in which $c(\cdot)$ is increasing and strictly convex (i.e., the production technology of an individual firm displays strictly decreasing returns to scale). We assume also that $x(c'(0)) > 0$. With this type of cost function, *no long-run competitive equilibrium can exist*. To see why this is so, note that if $p > c'(0)$, then $\pi(p) > 0$ and therefore the long-run supply is infinite. On the other hand, if $p \leq c'(0)$, then the long-run supply is zero while $x(p) > 0$. The problem is illustrated in Figure 10.F.2, where the graph of the demand function $x(\cdot)$ has no intersection with the

27. In particular, if (p^*, q^*, J^*) is a long-run equilibrium, then condition (i) of Definition 10.F.1 implies that $q^* \in q(p^*)$ and condition (iii) implies that $\pi(p^*) = 0$. Hence, by condition (ii), $x(p^*) \in Q(p^*)$. In the other direction, if $x(p^*) \in Q(p^*)$, then $\pi(p^*) = 0$ and there exists $q^* \in q(p^*)$ and J^* with $x(p^*) = J^*q^*$. Therefore, the three conditions of Definition 10.F.1 are satisfied.

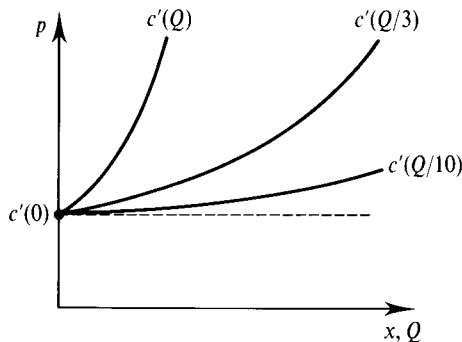
**Figure 10.F.2**

Nonexistence of long-run competitive equilibrium with strictly convex costs. (a) A firm's supply correspondence. (b) No intersection of long-run supply and demand.

graph of the long-run aggregate supply correspondence

$$Q(p) = \begin{cases} \infty & \text{if } p > c'(0) \\ 0 & \text{if } p \leq c'(0). \end{cases}$$

The difficulty can be understood in a related way. As discussed in Exercise 5.B.4, the long-run aggregate production set in the situation just described is convex but not closed. This can be seen in Figure 10.F.3, where the industry marginal cost function with J firms,

**Figure 10.F.3**

The limiting behavior of industry marginal cost as $J \rightarrow \infty$ with strictly convex costs.

$c'(Q/J)$, is shown for various values of J (in particular, for $J = 1$, $J = 3$, and $J = 10$). Note that as J increases, this marginal cost function approaches *but never reaches* the marginal cost function corresponding to a constant marginal cost of $c'(0)$.

Perhaps not surprisingly, to generate the existence of an equilibrium with a determinate number of firms, the long-run cost function must exhibit a strictly positive efficient scale; that is, *there must exist a strictly positive output level $\bar{q}$ at which a firm's average costs of production are minimized* (see Section 5.D for a further discussion of the efficient scale concept).

Suppose, in particular, that $c(\cdot)$ has a unique efficient scale $\bar{q} > 0$, and let the minimized level of average cost be $\bar{c} = c(\bar{q})/\bar{q}$. Assume, moreover, that $x(\bar{c}) > 0$. If at a long-run equilibrium (p^*, q^*, J^*) we had $p^* > \bar{c}$, then $p^* \bar{q} > \bar{c} \bar{q}$, and so we would have $\pi(p^*) > 0$. Thus, at any long-run equilibrium we must have $p^* \leq \bar{c}$. In contrast, if $p^* < \bar{c}$, then $x(p^*) > 0$; but since $p^* q - c(q) = p^* q - (c(q)/q)q \leq (p^* - \bar{c})q < 0$

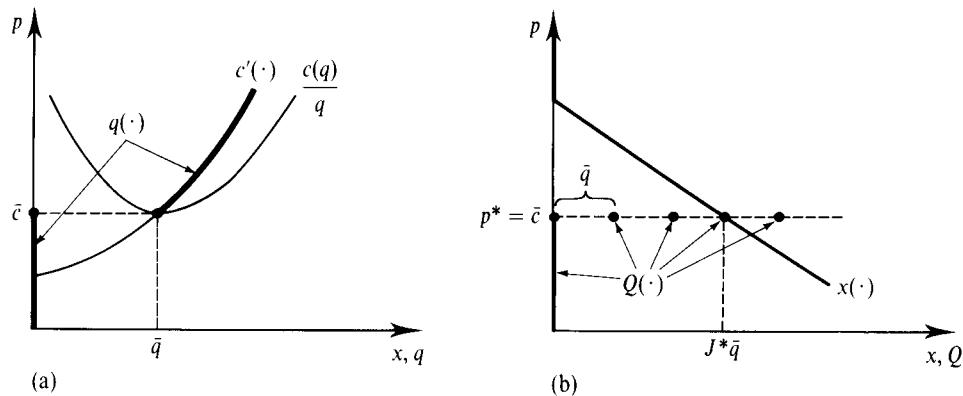


Figure 10.F.4
Long-run competitive equilibrium when average costs exhibit strictly positive efficient scale. (a) A firm's supply correspondence. (b) Long-run equilibrium.

for all $q > 0$, a firm would earn strictly negative profits at any positive level of output. So $p^* < \bar{c}$ also cannot be a long-run equilibrium price. Thus, at any long-run equilibrium we must have $p^* = \bar{c}$. Moreover, if $p^* = \bar{c}$, then each active firm's supply must be $q^* = \bar{q}$ (this is the only strictly positive output level at which the firm earns nonnegative profits), and the equilibrium number of active firms is therefore $J^* = x(\bar{c})/\bar{q}$.²⁸ In conclusion, the number of active firms is a well-determined quantity at long-run equilibrium. Figure 10.F.4 depicts such an equilibrium. The long-run aggregate supply correspondence is

$$Q(p) = \begin{cases} \infty & \text{if } p > \bar{c} \\ \{Q \geq 0: Q = J\bar{q} \text{ for some integer } J \geq 0\} & \text{if } p = \bar{c} \\ 0 & \text{if } p < \bar{c}. \end{cases}$$

Observe that the equilibrium price and aggregate output are exactly the same as if the firms had a constant returns to scale technology with unit cost $\bar{c}$.

Several points should be noted about the equilibrium depicted in Figure 10.F.4. First, if the efficient scale of operation is large relative to the size of market demand, it could well turn out that the equilibrium number of active firms is small. In these cases, we may reasonably question the appropriateness of the price-taking assumption (e.g., what if $J^* = 1$?). Indeed, we are then likely to be in the realm of the situations with market power studied in Chapter 12.

Second, we have conveniently shown the demand at price $\bar{c}$, $x(\bar{c})$, to be an integer multiple of $\bar{q}$. Were this not so, no long-run equilibrium would exist because the graphs of the demand function and the long-run supply correspondence would

28. Note that when $c(\cdot)$ is differentiable, condition (i) of Definition 10.F.1 implies that $c'(q^*) = p^*$, while condition (iii) implies $p^* = c(q^*)/q^*$. Thus, a necessary condition for an equilibrium is that $c'(q^*) = c(q^*)/q^*$. This is the condition for q^* to be a critical point of average costs [differentiate $c(q)/q$ and see Exercise 5.D.1]. In the case where average cost $c(q)/q$ is U-shaped (i.e., with no critical point other than the global minimum, as shown in Figure 10.F.4), this implies that $q^* = \bar{q}$, and so $p^* = \bar{c}$ and $J^* = x(\bar{c})/\bar{q}$. Note, however, that the argument in the text does not require this assumption about the shape of average costs.